

Causal Probabilistic Transformer: A Probabilistic Dependency Model for Autoregressive Language Modelling

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Abstract

The probabilistic transformer of Wu and Tu (2023) derives a transformer encoder from mean-field inference in a conditional random field over latent syntactic labels and dependency heads. We derive its autoregressive counterpart. Because causality, a single global energy and contextual dependence cannot hold together, we use a directed chain of conditional random fields in which the prefix enters each step as a frozen condition. Treating the next word as a latent variable of the same graph gives a language-model head with no parameter outside the graph, with tied embeddings and a closed-form mixture-of-softmaxes readout. At matched non-embedding budget on the Penn Treebank, its perplexity is 1.23 to 1.55 times that of a causal transformer. Much of its sensitivity to width comes from a miscalibrated attention temperature; correcting it with a normalisation outside the factor graph roughly doubles the usable label-set size and transfers to a second corpus without retuning. On the range where all three models are measured, the scaling exponents are -0.175 for a causal transformer, -0.068 for a looped transformer and -0.045 for our model, so most of its shallower slope already appears with weight sharing alone.

1 Introduction

Wu and Tu (2023) derive a transformer encoder from a probabilistic model. They define a conditional random field (CRF) over latent syntactic labels of words and dependency arcs between them, run mean-field variational inference (MFVI), and find that the resulting computation graph closely resembles a transformer encoder (Vaswani et al., 2017), with self-attention as the posterior over dependency heads. The model is meant to explain what a transformer might compute, not to

improve on it. That account covers only the encoder, while most transformers today are autoregressive decoders. We ask whether it survives the move to generation, and what it costs.

A causal mask on the dependency factors is not enough. Under a single smooth energy every coupling is two-sided, so masking one direction removes the other as well: causality, a single global energy and contextual dependence cannot hold together (Theorem 1). We give up the single energy and use a directed chain of CRFs, one per position, in which the prefix posteriors enter as a frozen condition, so the anti-causal term is never formed.

A decoder also needs a distribution over the vocabulary, and a projection into the vocabulary would have no factor behind it. The word, however, is already a variable of the graph, only observed; unclamping it turns the word-label factor into a language-model head. Tied input and output embeddings follow, and so does the output bias. Encoding and decoding become two conditioning patterns of one model. Given the prefix, the subgraph at each position is a tree, so exact inference gives a mixture-of-softmaxes readout in closed form. The mean-field readout has the softmax bottleneck of a transformer’s head at matched width, and the exact readout escapes it at no parameter cost (Proposition 4).

Our model, the causal probabilistic transformer (causal PT), differs from a standard decoder in two ways at once: it shares its parameters across inference iterations, and it has a syntactic factor graph. A looped transformer shares weights without the structure, so comparing with it isolates structure, and a set of ten single-difference ablations (the switch ladder) measures what each difference costs on its own. Much of the model’s sensitivity to width comes from the attention temperature of Wu and Tu, which is calibrated only at the uniform belief. Correcting it lets larger label sets train where they would otherwise collapse to the unigram dis-

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tribution. The best model reaches a validation perplexity of 201.9 on the Penn Treebank, against 445.2 for a bigram model. Across sizes its perplexity is 1.23 to 1.55 times that of a causal transformer interpolated to the same non-embedding parameter count. Without supervision, some labels of a small model align with syntactic categories, and in a wider model the heads prefer nearby words (mean attachment distance 11 against 17 under uniform attention); how much of either appears depends on the parameterisation (Appendix G).

2 Causal Probabilistic Transformers

We recall the model of Wu and Tu (2023), show why masking it is not enough, and build the causal model.

2.1 Background

A probabilistic transformer is a CRF over two families of discrete latent variables. Each position i of a sentence has a *label* variable Z_i over d values (the model’s *width*), read as a syntactic category, and h *head* variables $H_i^{(1..h)}$, one per *channel*, each over the candidate heads of i . A unary factor $\exp(S_{w_i, Z_i})$ ties the label at i to the observed word through one matrix $S \in \mathbb{R}^{|\mathcal{V}| \times d}$, and a ternary factor $\exp(\mathbf{1}[H_i^{(c)} = j] T_{Z_i, Z_j}^{(c)})$ scores the labels at i and at its head j in channel c .

Unrolled for T iterations, mean-field inference is scaled dot-product attention (the head update) followed by a re-injected unary (the label update): a T -layer transformer encoder with shared weights and no feed-forward block. The word w_i enters the model in one place only: the unary factor $S_{w_i, \cdot}$.

2.2 Why a causal mask is not enough

Masking the ternary factors so that $H_i^{(c)}$ can only point to positions $j < i$ does not give a causal model, because three properties cannot hold together. *Single global energy* (SE): the updates are $Q_x \propto \exp(-\nabla_x E / \lambda_x)$ for one twice continuously differentiable energy E , with the words entering only through terms local to a single position. *Causality* (C2): for every schedule and every node, the update at position i does not read the state of any node at a position $j > i$. *Contextuality* (X): some belief at some position changes when a word at another position changes.

Theorem 1 (Trilemma). *No scheme satisfies SE, C2 and X simultaneously. Any two of the three are realisable.*

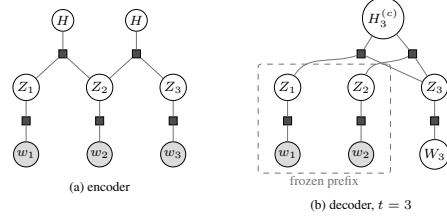


Figure 1: The same factor graph in its two conditioning patterns ($n = 3$); shaded nodes are observed. (a) Encoder: every word is clamped and arcs are undirected, as in Wu and Tu (2023). (b) Decoder at slot $t = 3$: prefix labels are frozen conditions (boxed), arcs point backwards, and the unclamped word w_3 carries the next-token distribution. Factors and parameters are the same in both panels.

If E is twice continuously differentiable, $\partial^2 E / \partial Q_i \partial Q_j$ is symmetric, so the term by which the label at j reaches the update at i and the reverse term are one Hessian entry. Causality deletes one of them and symmetry then deletes the other. The proof is in Appendix C. The theorem needs the architectural form of causality, C2, which masked attention satisfies; under a weaker trajectory form, C1, it fails (Proposition 5). It also covers only context that passes through inference: an energy with terms that read words at other positions could condition on the prefix directly. We therefore give up the single global energy.

Remark 2 (The anti-causal term). Deleting the anti-causal contribution from the symmetric model gives a field that is not the gradient of any energy, and by Theorem 1 no repair of it can be. In the chain below, the factor in which position i heads a later word j lives inside $p(Z_j, H_j | z_{<j})$, where z_i is a conditioning value, so that contribution never exists.

2.3 A directed chain of conditional CRFs

Instead of one CRF over the sentence we define a chain of conditional CRFs, one per position. The CRF at position i ranges over $(Z_i, H_i^{(1..h)})$ and is conditioned on the realised prefix labels $z_{<i}$:

$$p(Z_i, \{H_i^{(c)}\}_c | z_{<i}, w_{1:i}) \propto \exp \left(S_{w_i, Z_i} + \sum_c \sum_{j \in \mathcal{D}_i} \mathbf{1}[H_i^{(c)} = j] \tau_{Z_i}^{(c)}(j) \right), \quad (1)$$

with head domain $\mathcal{D}_i = \{\text{ROOT}, 1, \dots, i-1\}$, arc score $\tau_a^{(c)}(j) = T_{a, z_j}^{(c)}$ for $j < i$, and $\tau_a^{(c)}(\text{ROOT}) = r_a^{(c)}$ a learned root column. The joint is the chain-rule product $p(z, h | w) = \prod_i p(z_i, h_i | z_{<i}, w_{1:i})$,

a valid probability model by construction, and, as Theorem 1 requires, there is no single global energy. The root column gives the first word something to attend to, much as a beginning-of-sequence token does in a transformer.

Inference is variational filtering. We process $i = 1, \dots, n$ and keep frozen filtering marginals $\bar{q}_j \approx p(Z_j \mid w_{1:j})$ for $j < i$. Because the log-potential is linear in the one-hot encoding of z_j , its expectation under $\bar{q}_j$ is obtained by substituting $\bar{q}_j$ for that encoding:

$$\mathbb{E}_{\bar{q}_j} [T_{a,Z_j}^{(c)}] = \sum_b \bar{q}_j(b) T_{a,b}^{(c)} =: B_{j,a}^{(c)}. \quad (2)$$

Step i is therefore mean-field inference on a small CRF with unary $S_{w_i,\cdot}$ and binary factors $\exp(B_{j,a}^{(c)})$, with updates

$$\begin{aligned} Q_{H_i^{(c)}}(j) &\propto \exp\left(\frac{1}{\lambda_H} \sum_a Q_{Z_i}(a) B_{j,a}^{(c)}\right), \quad (3) \\ Q_{Z_i}(a) &\propto \exp\left(\frac{1}{\lambda_Z} [S_{w_i,a} + \mathcal{G}_i(a)]\right), \\ \mathcal{G}_i(a) &= \sum_c \sum_{j \in \mathcal{D}_i} Q_{H_i^{(c)}}(j) B_{j,a}^{(c)}. \quad (4) \end{aligned}$$

With $T^{(c)} = U^{(c)} V^{(c)\top}$ this is masked scaled dot-product attention: the query $Q_{Z_i} U^{(c)}$ comes from the current position and the keys $\bar{q}_j V^{(c)}$ from frozen prefix states, so the key-value cache follows from freezing the prefix. Frozen means constant for step i 's inference problem, not detached in automatic differentiation: training gradients flow through $B_{j,\cdot}^{(c)}$ into the prefix computation, as through cached activations in a causal transformer, and this is how S is trained in both of its roles. This affects only parameter learning: in the forward pass no belief at a position depends on a later word (Appendix E).

2.4 The output mechanism

Unclamping the word variable. We call the position whose word is being predicted the *slot*. A new $d \times |\mathcal{V}|$ projection of Q_{Z_t} would give next-token logits, but it would correspond to no factor, Q_{Z_t} describes the word already observed at t , and no factor links Z_t to the next word, since arcs connect a label only to its head's label. The only factor touching a word variable is the word-label factor, and the only factors touching a label are that one and its arcs, so any route from prefix to vocabulary that stays in the graph must be $\bar{q}_{<t} \xrightarrow{\text{arcs}} Z_t \xrightarrow{S} W_t$. Making each word a variable turns the chain into a joint over words,

$p(w, z, h) = \prod_t p(W_t, Z_t, H_t \mid z_{<t})$, with step conditional

$$\begin{aligned} p(W_t, Z_t, H_t \mid z_{<t}) &\propto \exp\left(\underbrace{b_{W_t}}_{\text{word unary}} + \underbrace{S_{W_t, Z_t}}_{\text{word-label factor}} \right. \\ &\quad \left. + \sum_c \sum_{j \in \mathcal{D}_t} \mathbf{1}[H_t^{(c)} = j] T_{Z_t, z_j}^{(c)}\right). \quad (5) \end{aligned}$$

No factor or parameter is added: the unary $\exp(S_{w_t, Z_t})$ of Section 2.1 is the binary factor $\exp(S_{W_t, Z_t})$ with W_t clamped to the observed word.

Consequences. (i) *Tying is forced.* The step conditional has one word-label factor, used both to condition on $W_t = w_t$ and to predict it. Separate matrices for the two roles would be two factors on the same pair of variables, and $\exp(S_{w,a}) \exp(S'_{w,a}) = \exp((S+S')_{w,a})$ collapses them into one. Untying gives a different model. (ii) *The output bias is the word unary b* , a per-word constant; in our experiments it is fixed to the log unigram frequency of the training split. (iii) *One model, two conditioning patterns.* We call the pattern in which W_t is clamped the *observed mode* and the one in which it is latent the *predictive mode*. The observed-mode energy is the predictive-mode energy with Q_{W_t} clamped to δ_{w_t} (Figure 1).

2.5 Inference at a slot, and the exact readout

At slot t , with the prefix frozen and contracted by Equation (2), the updates of Equations (3) and (4) gain a word update $Q_{W_t}(w) \propto \exp\left(\frac{1}{\lambda_W} [b_w + \sum_a Q_{Z_t}(a) S_{w,a}]\right)$, whose softmax is the language-model output layer. The word variable starts from its unary, so the initial query is a prior-weighted mean embedding that plays the role of a [MASK] token (Appendix A.1).

The slot's factor graph is a star centred on Z_t , with edges $W_t - Z_t$ and $H_t^{(c)} - Z_t$, hence a tree, so sum-product is exact and has a closed form.

Proposition 3. *For the slot conditional of Equation (5) with the prefix frozen,*

$$\begin{aligned} \hat{p}(W_t = w \mid w_{<t}) &\propto e^{b_w} \sum_a e^{S_{w,a}} \mu_t(a), \\ \log \mu_t(a) &= \sum_{c=1}^h \text{LSE}_{j \in \mathcal{D}_t} B_{j,a}^{(c)}, \quad (6) \end{aligned}$$

with the log-sum-exp seeded at $j = \text{ROOT}$ by $r_a^{(c)}$.

This is a mixture of d fixed softmax components with context-dependent weights, so it escapes the rank constraint of the mean-field readout

$\exp \mathbb{E}_{Q_{Z_t}}[S_{w,\cdot}]$ (Section 3.3). Since $\mathcal{D}_t$ grows by one position per step, one logcumsumexp scan computes it in $O(ndh)$, with no predictive inner loop (Appendix A.2). Every step is then valid inference in the declared model: mean field on the chain, where exact inference is intractable, and exact sum-product on the slot.

2.6 Variants

The clipped distance-sensitive arc score of Wu and Tu keeps only its $i - j > 0$ half, with $\gamma + 1$ buckets. All reported numbers use a layer-parallel schedule, which updates every position from the previous iteration under a strictly lower-triangular mask: a depth- T causal transformer with shared parameters (Appendix B.2).

Two variants add variables. Wu and Tu propose global variables as an in-graph substitute for the feed-forward block but report no results for them. A single-split global variable G_t , added as a leaf off Z_t , keeps the slot a tree and the model causal; its mean-field update is the feed-forward-like operator $\sigma(qB'^\top)B'$ (Appendix B.1). *Factored labels* replace the label variable by K variables of width d/K that share W_t ; the arc parameters fall with K (Appendix B.4). The slot is still a tree, and the exact readout becomes a product of K mixtures. Under the mean-field readout the logits stay affine in the concatenated beliefs, so at a fixed total label width we predict that factoring helps only through the exact readout.

Other extensions must also add variables: a multilayer perceptron on Q_{Z_t} or untied per-iteration parameters would add a map with no factor behind it.

3 Comparison with Causal Transformers

Wu and Tu (2023) show that MFVI in their model has the computation graph of a transformer encoder. The match is closer for the causal model: the factor of two from symmetrisation disappears, $T^{(c)}$ need not be symmetric because the arcs are directed, and excluding self-attachment ($j = i$) is harmless because the re-injected unary plays the residual role.

3.1 Tensor form

Collect the label beliefs of the prefix into $\bar{Q} \in \mathbb{R}^{n \times d}$ with rows $\bar{q}_j$, and write $Q_Z^{(t)} \in \mathbb{R}^{n \times d}$ for the beliefs after iteration t (here t indexes iterations, not positions). With $T^{(c)} = U^{(c)}V^{(c)\top}$ and

$U^{(c)}, V^{(c)} \in \mathbb{R}^{d \times r}$, Equations (3) and (4) become, for channel c ,

$$\begin{aligned}\mathcal{F}^{(c)} &= \frac{1}{\lambda_H} (Q_Z^{(t-1)} U^{(c)}) (\bar{Q} V^{(c)})^\top + M, \\ A^{(c)} &= \text{softmax}_{\text{rows}} (\mathcal{F}^{(c)}), \\ \mathcal{G} &= \sum_c A^{(c)} \bar{Q} V^{(c)} U^{(c)\top}, \\ Q_Z^{(t)} &= \text{softmax}_{\text{rows}} \left(\frac{1}{\lambda_Z} (S_{w,\cdot} + \mathcal{G}) \right),\end{aligned}\tag{7}$$

where M is the strict lower-triangular mask with the ROOT column prepended. $Q_Z^{(t-1)} U^{(c)}$ is the query projection, $\bar{Q} V^{(c)}$ the key projection, λ_H the attention temperature and $\mathcal{G}$ the attention output.

3.2 The update vs. masked self-attention

Projections, residual and normalisation.

$\bar{Q} V^{(c)}$ appears twice in Equation (7), as the key and as the value: both roles are the same message through the same arc factor. There is no output projection W^O either; the energy sums over channels, so their messages are summed. There is no residual stream and no layer normalisation. The unary $S_{w,\cdot}$, added again at every iteration, plays the residual role, and the simplex keeps activations bounded. Wu and Tu control the parameters with an L_2 penalty on the arc scores.

The temperature. Layer normalisation holds a transformer’s queries and keys at a roughly constant scale, so one temperature $1/\sqrt{d_{\text{head}}}$ serves throughout training. Here the query Q_{Z_i} and the key source $\bar{q}_j$ are probability vectors, whose Euclidean norm ranges over $[d^{-1/2}, 1]$ from uniform to one-hot, so the scale of the head logit $\langle Q_{Z_i}, B_{j,\cdot}^{(c)} \rangle$ can change by a factor of up to d during training. Wu and Tu set $\lambda_H = 1/d$ (the *original temperature*). With arc scores of scale σ_T , this gives a scaled head logit of $O(\sigma_T)$ at the near-uniform beliefs of initialisation but $O(d\sigma_T)$ once the beliefs sharpen, an error that grows linearly with the width. We therefore predict a positive feedback loop $\bar{q}_j \rightarrow B_{j,\cdot}^{(c)} \rightarrow \mathcal{G} \rightarrow \bar{q}_j$ whose gain grows with d , so that any damping of the label update must get stronger as the model gets wider.

We test two repairs. The *belief-norm temperature* $\lambda_H \|Q_{Z_i}\|_2 \sqrt{d}$ equals λ_H at the uniform belief and stays inside the model. The *standardised temperature* centres the head logits over $\mathcal{D}_i$, scales them to unit root mean square and multiplies them by a gain g (default 1). It makes attention invariant to the scale of the arc scores, but it is a normalisation outside the update equations.

3.3 The readout vs. a language-model head

The encoder has no output layer. Let L be the matrix whose rows are the log-conditionals $\log p(\cdot \mid \text{context})$ over the vocabulary for a fixed set of contexts.

Proposition 4 (Readout expressivity). (i) *Under the mean-field readout $\ell = b + SQ_{Z_t}$ with $Q_{Z_t} \in \Delta^{d-1}$, the rows of L lie in an affine subspace of dimension at most d .* (ii) *Under a linear-softmax head of width d_{model} they lie in an affine subspace of dimension at most $d_{\text{model}} + 1$.* (iii) *The exact readout of Equation (6) is not an affine family: unless the component distributions $p(\cdot \mid a) \propto e^{b+S_{\cdot,a}}$ are all identical, $\mu \mapsto \log \hat{p}(\cdot)$ is nonlinear in its parameters and its image is a curved $(d-1)$ -dimensional manifold not contained in the subspace of (i).*

The mean-field readout therefore has the softmax bottleneck of Yang et al. (2018), the same as a transformer’s output head at $d = d_{\text{model}}$. The exact readout is weaker than a full mixture of softmaxes, but it lies outside the linear family at no parameter cost. The prefix reaches the future as one point of Δ^{d-1} per position, where a transformer’s cache holds one point of $\mathbb{R}^{d_{\text{model}}}$.

3.4 Full model comparison

Figure 2 shows one layer of each model. The causal PT differs from the transformer in ten ways, each of which can be switched separately in the *switch ladder* (Section 4.3.4): parameters shared across iterations (weight_sharing), no feed-forward block (ffn), no output projection (attn_out_proj), keys tied to values (attn_value), no query projection (attn_query_proj), no residual stream (residual), no layer normalisation (norm), a simplex-constrained state (state), relative clipped positions (position), and a mixture readout over labels in place of a linear head (readout).

A looped transformer (Dehghani et al., 2019; Giannou et al., 2023; Yang et al., 2024) shares weights across depth, as the causal PT does, but has no syntactic structure. Comparing it with the transformer isolates weight sharing, and comparing the causal PT with it isolates structure.

4 Experiments

4.1 Tasks and Datasets

We train language models on the Penn Treebank (PTB; Marcus et al., 1993) and use WikiText-2 (Merity et al., 2017) as a transfer test only (Appendix D). Two references are computed on the

same scored tokens: the unigram of the training split (688.8 on PTB, 964.8 on WikiText-2) and an add-one bigram (445.2 and 692.5).

4.2 Settings

One training loop is shared by all models (Appendix D). As in Wu and Tu’s Table 2, we use AdamW with $\beta = (0.9, 0.999)$, weight decay 1.4×10^{-6} , gradient clipping at 1.0, 100 warm-up steps, cosine decay to $0.1\times$, and an L_2 penalty of 5×10^{-4} on the arc scores. Unless stated otherwise, every model trains for 15,000 steps on batches of 16 blocks of 64 tokens, without dropout; the width, gain, learning-rate, rank and initialisation grids use 6,000 steps. All models are scored on the same tokens at their best-validation checkpoint. Test perplexity at that checkpoint is 5 to 9 % below validation for every model in Tables 6 and 14 and orders the models the same way.

The baselines are nanoGPT decoders (Karpathy, 2022) with four layers and four heads; the looped transformer applies one such block four times. Each baseline point is the better of learning rates 10^{-3} and 3×10^{-3} . The causal PT uses $h = 2$, $T = 3$, $\gamma = 3$ and a learning rate of 2×10^{-2} , twenty times Wu and Tu’s (Appendix D). Its label update can be damped with a step $\alpha_Z \leq 1$ (Wu and Tu’s Appendix B.1). The scaling ladders run undamped under the standardised temperature at gain 1. The *damped configuration* (original temperature, $\alpha_Z = 0.25$, $d = r = 32$) is the base of the constants, depth and readout sweeps. All causal PT numbers use the mean-field readout except in Section 4.3.6; we did not retrain the scaling, width, rank and switch grids under the exact readout. With tying, almost all of the causal PT’s parameters are in S (at $d = 32$, 330,000 embedding against 16,448 other parameters), so we report non-embedding and total counts separately. A seed *collapses* if it finishes above 95 % of the unigram perplexity. Collapsed seeds are counted and left out of the mean, except in Table 6, whose means include them.

4.3 Results

4.3.1 Correctness and reproducibility

On a single slot the exact readout matches brute-force enumeration to 6.7×10^{-16} ; the free energy never increases along the inner loop, a test that catches a sign-flipped head update missed by every other check; and changing token $t + 1$ leaves the logits at step t bitwise unchanged (Appendix E).

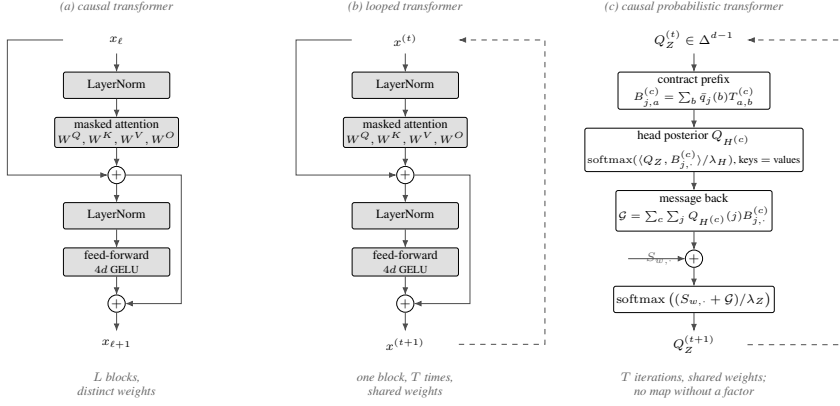


Figure 2: One layer of (a) a causal transformer, (b) a looped transformer, which applies the same block T times with shared weights, and (c) one MFVI iteration of the causal PT. Shaded boxes are learned maps with no factor behind them; (c) has none. In (a) and (b), d denotes d_{model} . Section 3.4 lists the ten differences.

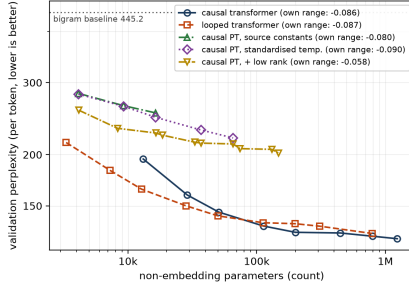


Figure 3: Validation perplexity on PTB against non-embedding parameters (log-log). Baseline points: better of two learning rates; causal PT points: three-seed means. ‘source constants’: original temperature, damped; ‘standardised temp.’: Table 6; ‘+ low rank’: Table 1. Legend slopes use each curve’s own range and are not comparable.

The undamped configuration at the original temperature ($d = 16$, 6,000 steps) is highly sensitive to last-bit perturbations: with the same code, seed and data order it ends at 430.0 on one GPU model and at 645.0 on another. Across three seeds on one device it gives 462.4 ± 130.2 (28.2 % of the mean); damping the label update ($\alpha_Z = 0.25$) reduces this to 318.4 ± 6.3 , 21 times tighter (Appendix H.1). We therefore report three-seed means with their spread and a count of collapsed seeds.

4.3.2 Scaling and matched budget

The transformer stops improving above about 2×10^5 non-embedding parameters, where it is limited by the size of PTB (Figure 3). The looped transformer beats it at matched non-embedding budget below about 5×10^4 parameters (Appendix H.2). On the range where all three models are measured (13k–148k non-embedding parameters), the exponents are -0.175 for the transformer, -0.068 for the looped transformer and -0.045 for the low-rank causal PT (Table 4). The looped model lies between the other two, closer to the causal PT, so

most of the slope difference already appears with weight sharing alone; the rest comes from the other nine differences of Section 3.4. The fits rest on four, three and eight points, so only this ordering is robust; pairwise fits, including the full-rank causal PT, give the same ordering in all five pairs (Table 5).

At matched non-embedding budget (Table 1) the gap widens with budget. The causal PT’s perplexity is 1.23 times the transformer’s at 16,512 parameters, the smallest budget with a measured transformer, and 1.53 times at 147,840; against the looped transformer the ratio runs from 1.25 to 1.48. The best causal PT reaches 201.9 ± 4.9 at $d = 192$, $r = 48$. These numbers use the standardised temperature, which is not part of the factor graph (see Limitations).

4.3.3 Width and the head temperature

Under the original temperature, damping is necessary and the need for it does not lift as d grows, consistent with the prediction of Section 3.2 (Figure 7). Undamped, the model ends on the unigram at $d \in \{32, 96, 128\}$, and removing the prefix changes its prediction by a KL divergence below 10^{-8} . The best damping step is the smallest we swept, $\alpha_Z = 0.125$, at five of seven widths, so the grid bounds the optimal step but does not locate it (Appendix H.3).

The standardised temperature revives dead widths without damping: at gain 1 and three seeds, $d = 32$ improves from 700.7 to 282.2 ± 6.6 and $d = 64$ reaches 254.0 ± 1.9 (Table 7, and Figure 8 for the other rules). At the full budget it gives 219.8 ± 4.2 at $d = 64$, where the damped control gives 470.0 ± 160.7 (Table 6). The correction roughly doubles the usable width but does not remove the limit: at $d \in \{96, 128\}$ the full-rank ladder is unstable again (372 ± 277 and 509 ± 187 ,

causal PT	non-emb.	val ppl	transformer	ratio	looped	ratio
$d = 32, r = 8$	4,160	256.8 ± 1.5	—	—	204.7	1.25
$d = 64, r = 8$	8,320	232.0 ± 1.6	—	—	178.5	1.30
$d = 48, r = 12$	9,312	235.5 ± 10.4	—	—	174.8	1.35
$d = 64, r = 16$	16,512	225.9 ± 2.5	184.1	1.23	159.8	1.41
$d = 96, r = 12$	18,624	223.1 ± 5.6	178.5	1.25	157.6	1.42
$d = 128, r = 16$	33,024	214.6 ± 2.0	155.9	1.38	147.7	1.45
$d = 96, r = 24$	37,056	213.5 ± 3.8	152.8	1.40	146.1	1.46
$d = 128, r = 32$	65,792	212.4 ± 4.0	141.2	1.50	139.9	1.52
$d = 192, r = 24$	74,112	206.9 ± 9.2	139.6	1.48	139.1	1.49
$d = 256, r = 32$	131,584	206.1 ± 4.8	132.6	1.55	136.2	1.51
$d = 192, r = 48$	147,840	201.9 ± 4.9	131.7	1.53	136.0	1.48
$d = 256, r = 64$	262,656	<i>1/3 collapsed</i>	129.1	—	134.5	—

Table 1: Matched-budget comparison on PTB along the causal PT’s low-rank scaling ladder (standardised temperature at gain 1, undamped, $h = 2$, $T = 3$, mean-field readout). Causal PT: mean $\pm$ sd over three seeds. Baselines are interpolated in log-log space to the causal PT’s non-embedding count and left blank outside their measured range; ratios are causal PT over baseline. *n/m collapsed*: n of m seeds ended on the unigram.

means that include one collapsed seed each). The best learning rate moves by at most one grid step over an eightfold range of width under either temperature (Table 8; one seed per cell), so the temperature seems to decide whether the model trains at all, a separate question from the learning-rate transfer of Kuang et al. (2026).

Rank. Rank equal to width is not the best choice: at $d = 96$, $r = 48$ gives 242.6 ± 3.3 against 275.6 ± 27.1 at $r = 96$ (Table 9), although at $d = 32$ full rank is best. Wu and Tu’s rule $r = \min(64, d)$ gives $r = d$ for every $d \leq 64$ (see Appendix H.3 for a likely reason). The low-rank ladder trains stably up to $d = 256$, $r = 32$ (206.1 ± 4.8), but at $d = 256$, $r = 64$ one seed in three still collapses: the ceiling moves without disappearing.

4.3.4 The switch ladder

The switch ladder measures the cost of each difference of Section 3.4 in two directions (Table 13; positive values mean the causal PT’s version is worse). S1 gives a tuned causal transformer one property of the causal PT; S2 restores one transformer property in a model with all ten properties of the causal PT (the all-PT model). In S1, value tying (-0.6), relative positions (-1.0) and dropping the query projection ($+1.9$) are nearly free, and so is layer normalisation once each switch has its own learning rate; the feed-forward block ($+15.1$) and weight sharing ($+12.5$) cost more, and with the mixture readout a transformer does not train at all (686.8). In S2, restoring nine of the ten transformer properties makes the all-PT model worse; only value tying is neutral ($+0.7$). With one seed per cell and the best of four learning rates per switch, differences of a point or two cannot be told from zero. Every S1 optimum is interior, but the all-PT model’s best learning rate lies at the top of its grid, so the S2

magnitudes are lower bounds (Appendix H.5).

4.3.5 Constants, global variables and depth

Under the standardised temperature at $d = 32$, a global variable has no effect (243.2 ± 5.3 without it, 242.5 to 254.1 with it across three initialisation scales, with and without the shared L_2 penalty); at $d = 64$ it changes which configurations are stable without making the model stable. Its posterior stays near uniform (entropy at least 0.93 of the maximum), so it sends a near-constant label prior. A semi-orthogonal initialisation of B' lowers the seed standard deviation from 10.2 (Gaussian) to 0.6 (Appendix H.4).

The constants and depth sweeps use the damped configuration and one seed per cell; its seed standard deviation is about 6 (Appendix H.6). The L_2 coefficient has an interior optimum near Wu and Tu’s value (5×10^{-4} and 5×10^{-3} are within noise), and of the clips tried (0, 1, 3, 7) only 1 and 3 train. More iterations do not help: $T = 1, 2, 3, 5$ give 241.3, 246.5, 248.3 and 297.0. The first three differ by about one standard deviation; $T = 5$ costs 56, about nine. $T = 8$ (275.3) breaks the trend, which one seed cannot resolve. The other experiments keep Wu and Tu’s $T = 3$.

4.3.6 Exact vs. mean-field readout, and factored labels

Swapping in the exact readout at evaluation, on mean-field-trained weights, multiplies perplexity by 7 to 18 across 11 configurations (Appendix H.7), so we train under each readout. With one seed each, the exact readout then reaches 348.8 against 248.3 at $d = 32$, and 409.5 against 315.5 at $d = 16$. It is the richer family (Proposition 4), yet it trains worse. We conjecture that its log-sum-exp over the head domain acts like a soft maximum and sends most of the gradient to the best-scoring position.

At a fixed arc budget of 8,256 parameters, going from $K = 1$ to $K = 2$ also doubles the label-set size, and the mean-field model improves from 283.8 ± 8.3 ($d = 16$) to 230.6 ± 4.4 ($d = 32$); $K = 4$ adds nothing. As predicted in Section 2.6, the exact readout’s penalty falls with K on both budgets we tested, from $+52.2$ at $K = 1$ to -3.5 at $K = 4$ on the smaller one (three seeds per cell; Table 15). The sign change in the last cell is within a seed spread, so the claim is the trend. At a fixed $d = 32$ the exact readout keeps improving while 94 % of the arc parameters are removed, but it never beats the mean-field readout’s best cell (230.6 at $K = 2$; Appendix H.8).

4.3.7 Transfer to WikiText-2

With nothing retuned, the standardised temperature beats the original one on WikiText-2 at both widths: 308.1 ± 2.6 against 343.1 ± 24.7 at $d = 32$, and 279.1 ± 9.0 against 334.1 ± 27.6 at $d = 64$ (Table 14). It also gains more from width, while the original temperature’s change is within its seed spread, which suggests the correction is not tuned to PTB. The gap to the baselines persists (Appendix H.9).

5 Related Work

Kuang et al. (2026) transfer maximal update parameterisation (Yang et al., 2021) to Wu and Tu’s encoder and find it ahead of a standard transformer at matched budget on masked language modelling up to 0.4B parameters, which suggests that small-scale gaps partly reflect tuning; our temperature correction is complementary to their rescaling. Yang et al. (2018) show that a linear-softmax head confines the log-conditionals to a low-dimensional affine family and propose a mixture of softmaxes; our exact readout is such a mixture, derived from inference on the slot tree and with no extra parameters. Tying input and output embeddings is a known improvement for language models (Press and Wolf, 2017); here it follows from the factor graph. Syntactic language models condition generation on explicit trees (Chelba and Jelinek, 1998; Dyer et al., 2016; Sartran et al., 2022) or on latent tree structure (Shen et al., 2019); our model keeps its syntax latent and never commits to a tree. Probes find some dependency structure in trained transformers (He-witt and Manning, 2019; Clark et al., 2019; Htut et al., 2019), depending on the objective and the probe (Ravishankar et al., 2021), and other work induces trees jointly with a language model (Shen

et al., 2021, 2022). Our head variable is an attachment by construction, and its clipped distance-sensitive arc scores play the role of the relative position representations of Shaw et al. (2018).

6 Discussion

What limits the model. At matched width the causal PT’s readout has a transformer’s bottleneck and its state has the same width, so what remains is the dynamics. The richer exact readout trains worse (Section 4.3.6), and in a one-seed sweep more iterations did not help (Section 4.3.5). This suggests that optimisation, not expressivity, is the limit at this scale. The exact readout’s penalty falls as the label variable is split (Section 4.3.6): with one label variable it may have had little extra to express.

Tuning and the feed-forward block. Much of the width sensitivity has one mechanism, the head temperature; the ceiling that remains after correcting it is unexplained (see Limitations). The failures at a large L_2 coefficient, at $\gamma = 7$ and under the original temperature share one cause: the ratio of the arc message to the word unary leaves the range in which the label update is stable (Appendix H.6); at $\gamma = 0$ the model cannot see distance. This ratio is worth monitoring while tuning, because perplexity stays flat until it jumps. The head temperature, learning rate, rank and label-set size matter most, and fewer iterations are better (an ordering from different budgets, not a ranking). The feed-forward block is the costliest single difference on the switch ladder that still trains, and the global variables proposed as its in-graph substitute do not close the gap because their posterior stays near uniform, a symmetry of the parameterisation that does not rule out other substitutes (Appendix H.4).

7 Conclusion

We derived a causal probabilistic transformer: a directed chain of conditional random fields with a frozen prefix, whose language-model head comes from unclamping the word variable. Its perplexity is 1.23 to 1.55 times a causal transformer’s at the same non-embedding budget, and most of the difference in scaling slope already appears with weight sharing alone. Correcting the head temperature roughly doubles the usable label-set size, while the exact readout and extra iterations did not help at this scale. Factored labels with the exact readout and fixed-lag smoothing are natural next steps.

Limitations

The width ceiling moves but does not lift. Correcting the head temperature roughly doubles the usable label-set size, but at $d \in \{96, 128\}$ one seed in three collapses. A lower rank of the arc scores moves the ceiling to $d = 256$ at $r = 32$, yet at $d = 256, r = 64$ one seed in three still collapses. We have no mechanism for this ceiling, and neither ladder should be extrapolated past the widths we measured.

The best repair is outside the graph. The standardised temperature is a normalisation outside the update equations, so the updates are no longer centred gradients of a fixed per-slot free energy. The belief-norm temperature stays inside the model but helps less (Figure 8). Our best numbers therefore come from a model that departs from the derivation in this one place; the label updates and the readout are unaffected.

Scale, tuning and baselines. All hyperparameters are selected on the Penn Treebank, which has under a million training tokens, and WikiText-2 serves only as a transfer check. On this corpus our baselines stop improving above about 2×10^5 non-embedding parameters, which favours the smaller causal PT. Wu and Tu (2023) report the encoder falling behind beyond about 10^5 sentences, which is why we did not use WikiText-103, while Kuang et al. (2026) find it ahead at matched budget once hyperparameters are transferred with maximal update parameterisation. We have not tested which pattern the causal model follows at scale. Without dropout and with a 64-token context, all models are far from published PTB perplexities; the comparison holds the protocol fixed and does not aim at the state of the art. The baselines also had a smaller tuning grid (two learning rates) than the causal PT.

Reproducibility. The undamped configuration at the original temperature is highly sensitive to last-bit perturbations: its seed standard deviation is 130.2, and a run that reproduces to the last printed digit on one GPU model ends 215 perplexity away on another (Section 4.3.1). Undamped configurations should be expected to differ on other hardware, and differences of tens of perplexity there should not be read as results.

What the switch ladder cannot say. The all-PT model at the end of the switch ladder approximates the causal PT in four respects (Appendix H.5),

chiefly a free projection into label space where the causal PT reads the label posterior directly. We cannot yet bound the gap between the all-PT model and the causal PT: it is about 77 perplexity at the best learning rate we ran, against about 170 at a single learning rate, and the all-PT model’s learning-rate sweep has not turned over. The ladder also runs four iterations and four channels, against three and two for the decoder, so part of this gap is the depth cost of Section 4.3.5. The single-switch (S1) estimates do not depend on this endpoint; the cumulative ladder should be read for its shape only.

The latent variables depend on the parameterisation. In the standardised configuration the head posterior prefers nearby heads, though half its mass lies beyond distance three; in the damped configuration it collapses onto the previous word (Appendix G). The labels behave the other way: most are readable at $d = 32$, but at $d = 96$, which gives the better perplexity, most are not. We found no setting where capacity and interpretability are both best.

No sequence-level joint. Giving up the global energy (Section 2.2) is what makes the model causal. Quantities defined by a sentence-level joint, such as a partition function or an exact sequence likelihood under bidirectional evidence, are therefore unavailable. The model is a valid directed joint and filtering is the right object for a decoder, but this is a real restriction relative to the encoder.

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A Derivations

A.1 Full mean-field updates

Fix a slot t . The prefix is frozen and contracted into the scores $B_{j,a}^{(c)}$ of Equation (2), with $B_{\text{ROOT},a}^{(c)} = r_a^{(c)}$. The live variables are Q_{W_t} on $\mathcal{V}$, Q_{Z_t} on the d labels, and one $Q_{H_t^{(c)}}$ on $\mathcal{D}_t$ per channel. The expected negative log of Equation (5) under their product gives, up to a constant, the per-slot energy

$$\begin{aligned} E_t = & -\sum_w Q_{W_t}(w) b_w \\ & -\sum_{w,a} Q_{W_t}(w) Q_{Z_t}(a) S_{w,a} \\ & -\sum_c \sum_{j \in \mathcal{D}_t} \sum_a Q_{H_t^{(c)}}(j) Q_{Z_t}(a) B_{j,a}^{(c)}. \end{aligned} \quad (8)$$

E_t is multilinear, so each coordinate step has a closed form. It is also one energy per slot. The prefix enters only through the constants $B_{j,a}^{(c)}$, and the factors in which slot t serves a later slot u belong to E_u , where $\bar{q}_t$ is a constant. No term is deleted, and the anti-causal term is never formed (Remark 2); this is how the construction respects Theorem 1. The gradients are

$$\frac{\partial E_t}{\partial Q_{W_t}(w)} = -b_w - \sum_a Q_{Z_t}(a) S_{w,a}, \quad (9)$$

$$\frac{\partial E_t}{\partial Q_{H_t^{(c)}}(j)} = -\sum_a Q_{Z_t}(a) B_{j,a}^{(c)}, \quad (10)$$

$$\begin{aligned} \frac{\partial E_t}{\partial Q_{Z_t}(a)} = & -\sum_w Q_{W_t}(w) S_{w,a} \\ & -\sum_c \sum_{j \in \mathcal{D}_t} Q_{H_t^{(c)}}(j) B_{j,a}^{(c)}. \end{aligned} \quad (11)$$

Inference minimises the free energy

$$\begin{aligned} \mathcal{F}_t = & E_t - \lambda_W H(Q_{W_t}) - \lambda_Z H(Q_{Z_t}) \\ & - \lambda_H \sum_c H(Q_{H_t^{(c)}}), \end{aligned} \quad (12)$$

the extended entropic Frank–Wolfe objective of Wu and Tu (2023) (their Appendix A.2), which is the ordinary mean-field free energy when all three weights are one. Each coordinate subproblem $\min_{z \in \Delta} \langle c, z \rangle + \lambda \sum_s z_s \log z_s$ is solved by $z = \text{softmax}(-c/\lambda)$ (their Appendix A.3), which gives the updates

$$Q_{H_t^{(c)}}(j) \propto \exp\left(\frac{1}{\lambda_H} \sum_a Q_{Z_t}(a) B_{j,a}^{(c)}\right), \quad (13)$$

$$\begin{aligned} Q_{Z_t}(a) \propto \exp\left(\frac{1}{\lambda_Z} \left[\sum_w Q_{W_t}(w) S_{w,a} \right. \right. \\ \left. \left. + \sum_c \sum_{j \in \mathcal{D}_t} Q_{H_t^{(c)}}(j) B_{j,a}^{(c)} \right] \right), \end{aligned} \quad (14)$$

$$Q_{W_t}(w) \propto \exp\left(\frac{1}{\lambda_W} \left[b_w + \sum_a Q_{Z_t}(a) S_{w,a} \right] \right). \quad (15)$$

A softmax ignores a constant shift, so only the centred gradient is read and the dropped constant has no effect. Equations (13) and (14) are Equations (3) and (4) with the word message $\sum_w Q_{W_t}(w) S_{w,a}$ in place of the observed unary. The softmax of Equation (15) over $\mathcal{V}$ is the output layer.

Schedule. Each variable is initialised from its own unary-type factors (Equation 7 of Wu and Tu): $Q_{W_t}^{(0)} \propto \exp(b)$ and $Q_{Z_t}^{(0)} \propto \exp(\tilde{s}/\lambda_Z)$ with $\tilde{s}_a = \sum_w Q_{W_t}^{(0)}(w) S_{w,a}$, the prior-weighted mean embedding that plays the role of a [MASK] token (Section 2.5). We then update $\{Q_{H_t^{(c)}}\}$ and Q_{Z_t} for ρ rounds in their asynchronous order (their §2.3.2) and read out with Equation (15). At $\rho = 1$ the attention query is the fixed learned probe $\text{softmax}(\tilde{s}/\lambda_Z)$; with $\rho \geq 2$ it depends on the context. Further $W \leftrightarrow Z$ rounds are valid mean field but feed the model’s own word guess back into the label belief, so we treat them as an ablation.

The observed mode is the same energy. Clamping Q_{W_t} to the point mass δ_{w_t} makes Equation (15) idle and reduces Equation (8), up to the constant $-b_{w_t}$, to

$$\begin{aligned} E_t^{\text{obs}} = & -\sum_a Q_{Z_t}(a) S_{w_t,a} \\ & -\sum_c \sum_{j \in \mathcal{D}_t} \sum_a Q_{H_t^{(c)}}(j) Q_{Z_t}(a) B_{j,a}^{(c)}, \end{aligned} \quad (16)$$

the per-position energy of the chain in Section 2.3, whose updates are Equations (3) and (4). Encoding and decoding are thus one energy family under two conditioning patterns (Figure 1), with S trained in both roles. By Equation (10) the head gradient contains no word message, so the attention update is the same function in both modes. Sharing one attention pattern between the predictive and the observed mode at a slot is therefore a choice of initialisation, which mean field leaves free.

A.2 Derivation of the exact readout

With the prefix frozen, the live factors at slot t are the word unary b , the word–label factor S and one contracted arc factor per channel. They form a star centred on Z_t , with leaves W_t and $H_t^{(1)}, \dots, H_t^{(h)}$. A star is a tree, so one sweep of sum–product gives the exact marginal. Each head variable appears in one term and has no unary, so eliminating it is a

plain sum over $\mathcal{D}_t$, and the channels separate:

$$\sum_{j_1, \dots, j_h \in \mathcal{D}_t} \exp \left(\sum_c B_{j_c, a}^{(c)} \right) = \prod_c \sum_{j \in \mathcal{D}_t} e^{B_{j, a}^{(c)}} =: \mu_t(a). \quad (17)$$

In logarithms, channel c sends $\text{LSE}_{j \in \mathcal{D}_t} B_{j, a}^{(c)}$ to Z_t , seeded at $j = \text{ROOT}$ by $r_a^{(c)}$. Eliminating Z_t leaves

$$\hat{p}(W_t = w \mid w_{< t}) \propto \sum_a e^{b_w + S_{w, a}} \mu_t(a) = e^{b_w} \sum_a e^{S_{w, a}} \mu_t(a), \quad (18)$$

which is Equation (6). The implementation works with the logits $b_w + \text{LSE}_a(S_{w, a} + \log \mu_t(a))$ and never forms μ_t itself. With $\mathcal{Z}_a = \sum_w e^{b_w + S_{w, a}}$ and $p(w \mid a) = e^{b_w + S_{w, a}} / \mathcal{Z}_a$, Equation (18) reads $\hat{p}(\cdot) = \sum_a \pi_t(a) p(\cdot \mid a)$ with mixture weights $\pi_t(a) \propto \mu_t(a) \mathcal{Z}_a$. The d components are fixed and only the weights depend on the context. This makes the readout a mixture of softmaxes (Section 3.3), weaker than one whose components also depend on the context.

The sequence form. Since $\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{t\}$, the channel message of Equation (17) is a causal prefix log-sum-exp in t . One logcumsumexp scan along the positions computes it for all t in $O(n dh)$, against $O(n^2 dh)$ slot by slot. Clipped distances keep this form. The causal model keeps the half $t - j > 0$ of the clip function of Wu and Tu (2023): distances $1, \dots, \gamma$ each have a bucket and longer distances share the last, so there are $\gamma + 1$ buckets. Writing $B_{j, a}^{(c, k)} = \sum_b \bar{q}_j(b) T_{a, b}^{(c, k)}$ for the contraction against the arc matrix of bucket k ,

$$\log \mu_t^{(c)}(a) = \text{LSE} \left(r_a^{(c)}, \text{LSE}_{1 \leq j \leq t - \gamma - 1} B_{j, a}^{(c, \gamma + 1)}, B_{t - 1, a}^{(c, 1)}, \dots, B_{t - \gamma, a}^{(c, \gamma)} \right), \quad (19)$$

where $\log \mu_t(a) = \sum_c \log \mu_t^{(c)}(a)$ and a term whose position index falls below 1 is $-\infty$. Only the far bucket accumulates, as one logcumsumexp offset by $\gamma + 1$. Each near bucket holds one position per slot and is its own table shifted by its distance, so the readout is one log-sum-exp over $\gamma + 2$ aligned tensors. The global variable of Appendix B.1, one more leaf off Z_t , adds the term $\text{LSE}_k B'_{k, a}$ to $\log \mu_t(a)$.

The layer-parallel schedule. The exact readout keeps this schedule. The readout is computed in closed form from the final prefix beliefs, the scan is an associative prefix reduction over $j < t$, and

there is no predictive inner loop, so a layer stays one triangular-masked pass (Section 2.6). At generation time the readout caches a running $d \times h$ table of channel messages, extended by one logaddexp per generated token.

A.3 Message weights

The weights $\lambda_Z, \lambda_H, \lambda_W$ of Equation (12), and λ_G for the global variable of Appendix B.1, each weight the entropy of one variable. Changing one changes the objective, so a sweep over λ_H is a sweep over objectives, not over a temperature within one objective.

λ_Z and λ_H . Wu and Tu (2023) fix $\lambda_Z = 1$ and derive $\lambda_H = 1/d$ from a variance argument (their Appendix A.5) in the manner of Vaswani et al. (2017). If the arc scores are independent with mean zero and variance σ^2 , the head logit $\sum_{a, b} Q_{Z_t}(a) \bar{q}_j(b) T_{a, b}^{(c)}$ of Equation (13) has variance σ^2/d^2 at uniform beliefs, and dividing by $\lambda_H = 1/d$ restores σ^2 . The causal head logit has the same form, with one belief frozen, so $\lambda_H = 1/d$ carries over. For λ_Z their argument gives no value: the variance of the label message depends on the sentence length, so they set $\lambda_Z = 1$. In the causal model the label message sums over $\mathcal{D}_t$, whose size is t (our extension of their argument), so no fixed λ_Z can restore the variance; we set $\lambda_Z = 1$. The encoder’s factor of two from symmetrising over head–child order is absent, since the directed model traverses each arc once.

λ_W . The word weight has no counterpart in Wu and Tu’s encoder, and none in the exact readout, which eliminates W_t exactly. Under the mean-field readout, tying lets the scale of S set both the output logits and the unary, so a value $\lambda_W \neq 1$ cannot be absorbed into S and would temper the readout. We use $\lambda_W = 1$ throughout; at any other value the evaluation-time swap of Section 4.3.6 would measure a temperature as well as the approximation.

λ_G . No argument fixes the weight of the global variable, and Wu and Tu write its update (their Appendix B.3) without one. We set $\lambda_G = 1$ without derivation and did not sweep it.

Calibration. The variance argument holds at uniform beliefs, where the model starts but does not stay. Section 3.2 analyses the consequence.

Layer	Width term	Length term
Transformer layer	$12d_{\text{model}}^2$	$2nd_{\text{model}}$
PT layer, observed mode	$3d^2$	$2nd$
PT, mean-field, both modes	$5d^2$	$4nd$
PT, exact readout	$3d^2$	$2nd$

Table 2: Approximate FLOPs per token and per layer, with $rh \approx d$ and $d = d_{\text{model}}$. A transformer’s $12d_{\text{model}}^2$ is $4d_{\text{model}}^2$ for the query, key, value and output projections plus $8d_{\text{model}}^2$ for the feed-forward block. The causal PT’s $3d^2$ is $3drh$ (prefix keys, which are also values, the query contraction and the output contraction). The exact-readout row adds, per token, $O(dh)$ for the scan and $O(|\mathcal{V}|d)$ for the vocabulary readout.

A.4 Complexity

Table 2 counts one layer at one position. A causal PT layer costs roughly half a transformer layer, because it has no feed-forward block and no projections. Running the mean-field readout, which solves a second inference problem in predictive mode at every slot, costs about $0.4\text{--}0.6\times$ a transformer layer. Global variables add about $2md$ per token. The vocabulary readout is an $O(|\mathcal{V}|d)$ log-sum-exp instead of a matrix product, with the same FLOPs and worse hardware constants. The implementation chunks it over the vocabulary, as a fused cross-entropy does, so the $|\mathcal{V}| \times d$ intermediate is never held in full.

Because the causal PT shares its parameters across iterations, depth and parameters cannot both be matched against a transformer. FLOPs also do not predict wall-clock time. Wu and Tu (2023) measure their encoder at about three times slower than a transformer despite far fewer parameters, and attribute this to gathering the arc matrices from the clipped distance table for every position pair (their Appendix C).

B Additional Variants

B.1 Global variables

Wu and Tu (2023) introduce global variables (their Appendix B.3) as an in-graph substitute for the feed-forward block. Of their three designs we use *single-split*, with one global variable per position shared by all channels (the others are *all-dep* and *dep-split*), which they identify as the closest to a transformer’s computation.

The construction is position-local, so it carries over unchanged. Slot t gains a variable G_t over m values and one binary factor $\phi_b(G_t=k, Z_t=a) = \exp(B'_{k,a})$ with $B' \in \mathbb{R}^{m \times d}$, so Equation (8) gains

the term $-\sum_{k,a} Q_{G_t}(k) Q_{Z_t}(a) B'_{k,a}$. There is no channel axis, so the message enters once per iteration. The slot is still a tree (G_t is one more leaf off Z_t), so Proposition 3 still applies; G_t reads no other position, so causality is untouched; and the cost is $O(md)$ per position. The mean-field updates gain one pair of messages,

$$Q_{G_t}(k) \propto \exp\left(\frac{1}{\lambda_G} \sum_a Q_{Z_t}(a) B'_{k,a}\right), \quad (20)$$

$$g_t^{\text{glob}}(a) = \sum_k Q_{G_t}(k) B'_{k,a},$$

the second of which joins the arc messages inside the bracket of Equation (4). Together they form the operator $\sigma(qB^T/\lambda_G)B'$ on the label belief q , which Wu and Tu identify as the analogue of a feed-forward block: a map into m features, a softmax in place of a RELU, and a map back through the transpose of the same matrix. Wu and Tu write this operator without a message weight; we carry λ_G , and $\lambda_G = 1$ recovers their form.

Under the exact readout the global variable acts only as a prior. Folding G_t into the sum-product of Proposition 3 multiplies $\mu_t(a)$ by $\sum_k e^{B'_{k,a}}$:

$$\log \mu_t(a) = \sum_c \text{LSE}_{j \in \mathcal{D}_t} B_{j,a}^{(c)} + \text{LSE}_k B'_{k,a}. \quad (21)$$

The added term is one d -vector, independent of t and of the prefix: a fixed per-label tilt of the mixture weights. It does not cancel, since the readout takes a log-sum-exp over labels, but it cannot carry context. The variable still reshapes every $\bar{q}_i$ through the observed-mode beliefs, so under the exact readout, which has no predictive inner loop, a run would mix a constant tilt with an indirect effect through the prefix. Our implementation refuses this combination, and validation checks assert that the direct term is constant and the end-to-end effect is not (Appendix E). Section 4.3.5 measures the mean-field configuration, in which the variable also acts at the predicted slot.

B.2 Serial update schedule

Both schedules are causal. The *serial* schedule runs Equations (3) and (4) at position i for a fixed number of inner rounds, freezes $\bar{q}_i$ and moves right. Each slot is then a well-posed mean-field problem on its own energy, with the per-step guarantees of the entropic Frank–Wolfe scheme, and causality follows by induction on i . The *layer-parallel* schedule computes all $Q_{Z_i}^{(t)}$ from $\{Q_{Z_j}^{(t-1)}\}_{j < i}$ at once under a strict lower-triangular mask, for T iterations.

Its computation graph is that of a depth- T causal transformer with shared weights. All reported numbers use this schedule, because only under it do the causal PT and the looped transformer share a computation graph and an effective depth, so that their difference isolates structure.

The two schedules do not compute the same thing. We compared their observed-mode label beliefs at $d = 32$, $h = 4$, $r = 8$, $\gamma = 3$, $T = 5$ and length 24. The mean total-variation distance is 3.4×10^{-6} at initialisation scale 0.02, 0.111 at 0.5 and 0.682 at 2.0, where the maximum reaches 1.000 (disjoint label mass at some positions). On an overfitted model it falls below 10^{-4} . The schedules agree when beliefs are near uniform or near one-hot, and can disagree completely in between.

B.3 Root node

The root is not a variable in this model. The head domain is $\mathcal{D}_t = \{\text{ROOT}, 1, \dots, t-1\}$, and $r^{(c)} = B_{\text{ROOT},\cdot}^{(c)}$ is a learned column of d numbers per channel. Like a beginning-of-sequence token, it gives position 1, whose head domain would otherwise be empty, something to attend to. The root of Wu and Tu (2023) (their §2.3.5) is an optional latent variable with its own label set and ternary potential. Ours is a constant parameter: it has no posterior, carries no information between positions and so cannot break causality, and costs hd numbers instead of a $d \times d_{\text{ROOT}} \times h$ tensor. It is their root variable with its distribution held fixed and contracted into a binary factor as in Equation (2), the collapse argument Wu and Tu use for global feature variables (their Appendix B.3.1).

At initialisation $r^{(c)}$ enters the attention in raw d -space, while the prefix rows arrive contracted, $B_{j,a}^{(c)} = \mathbb{E}_{\bar{q}_j}[T_{a,\cdot}^{(c)}]$, shrunk by roughly $d^{-1/2}$ for a near-uniform belief and again by the low-rank factorisation $T^{(c)} = U^{(c)}V^{(c)\top}$. With all parameters drawn from one $\mathcal{N}(0, \sigma^2)$, the ROOT row has $121 \times$ the norm of the prefix rows at $d = 384$, $h = 16$, $r = 64$. A near-uniform Q_{Z_t} hides this at first, but it could draw most of the attention once beliefs sharpen. In our measurements it does not: untrained, in the configuration above, the root receives $1.10 \times$ uniform attention mass at the last slot, and on an overfitted toy model it receives 0.0001 against 0.125 under uniform attention. How much mass a trained language model sends to the root still depends on the configuration (Appendix G). The initialisation scale of $r^{(c)}$ is a separate hyperpa-

rameter, left equal to that of the other parameters.

B.4 Factored labels

Factored labels replace the label variable at each position by K variables. As elsewhere, d is the total label width, so each component ranges over $d' = d/K$ values. Position t carries $Z_t^{(1)}, \dots, Z_t^{(K)}$ and K word-label factors that share the word variable, $\phi(W_t, Z_t^{(k)}) = \exp(S_{W_t, Z_t^{(k)}}^{(k)})$. Each channel c has a child component $k(c)$, whose belief queries the channel and receives its message, and a head component $k'(c)$, whose prefix belief forms the channel's keys: $B_{j,a}^{(c)} = \sum_b \bar{q}_j^{(k'(c))}(b) T_{a,b}^{(c)}$. The per-slot energy becomes

$$\begin{aligned} E_t = & - \sum_w Q_{W_t}(w) b_w \\ & - \sum_k \sum_{w,a} Q_{W_t}(w) Q_{Z_t}^{(k)}(a) S_{w,a}^{(k)} \\ & - \sum_c \sum_{j \in \mathcal{D}_t} \sum_a Q_{H_t^{(c)}}(j) Q_{Z_t}^{(k(c))}(a) B_{j,a}^{(c)}, \end{aligned} \quad (22)$$

still multilinear, so the updates are still softmaxes of the gradients. Causality is untouched, the attention correspondence holds per channel with component-specific queries and keys, and the compute is linear: K components of width d' cost what one of width d costs, since each channel touches one component. The slot graph is still a tree (W_t joined to K stars), so Equation (6) becomes a product of K mixtures,

$$\begin{aligned} \hat{p}(w) & \propto e^{b_w} \prod_{k=1}^K \sum_{a=1}^{d'} e^{S_{w,a}^{(k)}} \mu_t^{(k)}(a), \\ \log \mu_t^{(k)}(a) & = \sum_{c: k(c)=k} \text{LSE}_{j \in \mathcal{D}_t} B_{j,a}^{(c)}, \end{aligned} \quad (23)$$

a strictly richer family than one mixture at the same total width.

Under the mean-field readout, factoring adds no expressivity over a flat label of width d . The logits $b + \sum_k S^{(k)} q^{(k)}$ are affine in the concatenated beliefs, so their affine dimension is at most d , the bound of Proposition 4(i) for $K = 1$. Any gain in the head must come through Equation (23). Factoring also changes the inference dynamics and the parameter count, so Section 4.3.6 tests the interaction: the prediction is an effect of K under the exact readout that is absent under the mean-field readout, not an effect of the same size under both.

Channel assignment. We use $k(c) = c \bmod K$ and $k'(c) = \lfloor c/K \rfloor \bmod K$, so that at $h = K^2$ every ordered pair of components occurs exactly

once. The alternative $k'(c) = k(c)$, in which components never read each other, would separate the effect of having more mixture components from the effect of letting components communicate. We do not combine global variables with factored labels.

Parameter count. The word-label factors keep $|\mathcal{V}| K d' = |\mathcal{V}| d$ parameters, as for a flat label of width d ; S stays one tensor, read as K blocks of width d' . The arc factors, however, act on $d' \times d'$. The non-embedding block, $2(\gamma+1)h d' r$ for the arc scores plus $h d'$ for the root column, therefore falls as $1/K^2$ when the rank scales with d' and as $1/K$ when it is fixed. At $h = 4$, $\gamma = 3$ and $r = d'$ the measured non-embedding counts are 32,896 at $(d, K) = (32, 1)$, 8,256 at $(32, 2)$, 32,896 at $(64, 2)$, 8,256 at $(64, 4)$ and 32,896 at $(128, 4)$. Factoring therefore never gains through extra parameters; at a fixed non-embedding budget it gains total label width: the three equal counts are models of one size at $d = 32, 64$ and 128 .

C Proof of the Trilemma

Two formalisation choices come first.

Where the words enter. One might let each word enter only through its initialisation $Q_{z_i}^{(0)} \propto \exp(S_{w_i, \cdot})$. In the encoder model it also enters elsewhere: w_i appears in the unary term of E , so $S_{w_i, \cdot}$ reappears in every update of Z_i . We therefore assume only *word-locality* (A3 below); the initialisation-only version is the case $u_i \equiv 0$. The assumption matters. With word-nonlocal terms the theorem is false, as

$$E(Q; w) = - \sum_i \sum_a Q_i(a) \left(S_{w_i, a} + \sum_{k < i} S'_{w_k, a} \right) \quad (24)$$

shows. This is a single smooth energy that is causal and contextual, because it conditions on the prefix directly and bypasses inference. The theorem is about context mediated by inference over the marginals, and (A3) states that scope formally.

What causality means. (C1) *Trajectory causality*: for all $i < j$, all schedules and all t , the iterate $Q_{z_i}^{(t)}$, as a function of the words, does not depend on w_j . (C2) *Network causality*: for every schedule, every t and every node x , head nodes included, $Q_x^{(t)}$, as a function of an arbitrary interior initialisation, does not depend on the initialisation blocks of nodes at strictly later positions. The theorem is false under (C1). A finite vocabulary makes (C1) a

finite family of conditions on the states the trajectories actually visit, whereas the graph condition the proof needs is an identical-vanishing condition on the whole interior of the state space.

Proposition 5 (Sharpness). *Some scheme satisfies SE, C1 and X simultaneously.*

Proof. Take $n = 2$, binary labels ($d = 2$) and no head variables (permitted: the factor graph may have none, or only inert position-local ones). Let $\iota(v) = \text{softmax}(S_{v, \cdot})$ and $\Delta(v) = \iota(v)_1 - \iota(v)_2$, with the $\iota(v)$ and the $\Delta(v)$ pairwise distinct over the finite vocabulary, which is generic in S . With $\lambda = 1$ put

$$E(Q; w) = \psi(Q_1) \varphi(Q_2) - \sum_{i=1}^2 \langle S_{w_i, \cdot}, Q_i \rangle, \quad (25)$$

where $\psi(Q_1) = h(Q_1(1) - Q_1(2))$, $\varphi(Q_2) = Q_2(1)$, and h is the univariate Hermite interpolation polynomial with $h(\Delta(v)) = c_v$ pairwise distinct and $h'(\Delta(v)) = 0$ at every v in the vocabulary; such an h exists, of degree at most $2|\mathcal{V}| - 1$. Then E is polynomial, hence C^2 , and its pair term is word-free: SE holds. C1: whenever z_1 is updated its centred field is $\Pi_{z_1}[-S_{w_1, \cdot} + \varphi(Q_2) \nabla_{Q_1} \psi]$, and along the tangent direction $(1, -1)$ the second term contributes $2h'(\Delta(w_1)) = 0$ at the current state, so the update returns $\text{softmax}(S_{w_1, \cdot}) = \iota(w_1)$. The reachable set of z_1 is that fixed point under every schedule, and $Q_1^{(t)}$ never sees w_2 . X: under the schedule $B_1 = \{z_2\}$ the field driving z_2 contains $-\psi(Q_1^{(0)}) \Pi_{z_2}(1, 0)^\top = -c_{w_1} \Pi_{z_2}(1, 0)^\top$ with the c_v pairwise distinct, so $Q_2^{(1)}$ varies with w_1 , from past to future, the useful direction. C2 fails, as it must: off the finite critical set of h we have $h' \neq 0$ and the cross-position edge fires. The anti-causal coupling exists in E but is never reached. $\square$

C1 holds there only because h is fitted to the rows of S , and perturbing S with h fixed breaks it. C2 is what masked attention satisfies for all parameters, activations and inputs.

Setup. $\mathcal{N}$ has a node z_i per label variable Z_i and a node h_{ic} per head variable $H_i^{(c)}$, with $\pi(x)$ the position of x . Node x carries a distribution Q_x on a finite domain of size $d_x (= d$ for label nodes; for a head node the number of candidate heads, which the theorem leaves arbitrary, not necessarily causal). Write $\mathcal{X} = \prod_x \Delta_x$ and $\mathcal{X}^\circ = \prod_x \Delta_x^\circ$, $\mathcal{T}_x = \{u \in \mathbb{R}^{d_x} : \mathbf{1}^\top u = 0\}$, and $\Pi_x = I - d_x^{-1} \mathbf{1} \mathbf{1}^\top$ for the orthogonal centring projector onto $\mathcal{T}_x$. A function on $\mathcal{X}^\circ$ does

not depend on the y -block if it is constant under changes of the Q_y coordinates alone. Four assumptions hold throughout. (A1) $E \in C^2(U)$ for some open convex $U \supseteq \mathcal{X}$. (A2) A schedule is a sequence $(B_1, B_2, \dots)$ of non-empty subsets of $\mathcal{N}$; at step t each $x \in B_t$ is replaced by $Q_x^{(t)} = \text{softmax}(-\lambda_{x,t}^{-1} \nabla_x E(Q^{(t-1)}))$, $\lambda_{x,t} > 0$, and the rest carry over. This covers synchronous, asynchronous and mixed schedules and per-node, per-step message weights. (A3) $E(Q; w) = E_0(Q) + \sum_i u_i(Q_{z_i}; w_i)$ with E_0 word-free and u_i a function of the block Q_{z_i} alone (in the encoder, $u_i = -\langle S_{w_i, \cdot}, Q_i \rangle$), and nothing else depends on w . (A4) $Q_{z_i}^{(0)} = \iota_i(w_i) \in \Delta^\circ$, with head initialisations word-independent and interior. SE is (A1)–(A3) for one common E . X is the existence of an i , a t , a schedule and words differing only at some coordinate $k \neq i$ for which $Q_{z_i}^{(t)}$ differs.

Interiority: every iterate lies in $\mathcal{X}^\circ$, since initialisations are interior by (A4) and softmax outputs are strictly positive, so every gradient below is evaluated at an interior point. (A1) is needed and is not a technicality. The argument runs through equality of the mixed partials of E , which their existence alone does not give: for $f(x, y) = xy(x^2 - y^2)/(x^2 + y^2)$ with $f(0, 0) = 0$ both exist at the origin and disagree.

Gauge reduction.

Lemma 6. Fix a node x and $\lambda > 0$. For $P, P' \in \mathcal{X}^\circ$ the updates $\text{softmax}(-\nabla_x E(P)/\lambda)$ and $\text{softmax}(-\nabla_x E(P')/\lambda)$ agree if and only if $\Pi_x \nabla_x E(P) = \Pi_x \nabla_x E(P')$; so for $y \neq x$ the x -update depends on the y -block somewhere on $\mathcal{X}^\circ$ if and only if the centred field $\Pi_x \nabla_x E$ does.

Proof. $\text{softmax}(g) = \text{softmax}(g + c\mathbf{1})$ for every $c \in \mathbb{R}$, and softmax restricted to $\mathcal{T}_x$ is injective: $\text{softmax}(g) = \text{softmax}(g')$ forces $g - g' = c\mathbf{1}$, which centring forces to zero. Multiplication by $-1/\lambda$ is a bijection of $\mathcal{T}_x$. $\square$

An update therefore reads only the tangential part of the gradient. This closes the shift-invariance loophole of the softmax by definition: a coupling invisible to x 's softmax is one that, on the feasible set, does not depend on the x -block at all. Write $x \triangleright y$ when $\Pi_x \nabla_x E$ depends on the y -block somewhere on $\mathcal{X}^\circ$, and let Γ be the graph on $\mathcal{N}$ with an edge $\{x, y\}$ exactly when $x \triangleright y$.

Lemma 7 (No one-way coupling). For $x \neq y$, the conditions $x \triangleright y$; $u^\top \nabla^2 E(Q)v \neq 0$ for some

$u \in \mathcal{T}_x$, $v \in \mathcal{T}_y$ and $Q \in \mathcal{X}^\circ$; and $y \triangleright x$ are equivalent. In particular $\triangleright$ is symmetric and Γ is a well-defined undirected graph.

Proof. If $u^\top \nabla^2 E(Q)v \neq 0$ then $Q + sv \in \mathcal{X}^\circ$ for $|s|$ small, one block moving inside its open simplex, and $g(s) = u^\top \Pi_x \nabla_x E(Q + sv) = u^\top \nabla_x E(Q + sv)$ has $g'(0) \neq 0$. It is non-constant along a segment whose endpoints differ only in the y -block, so $x \triangleright y$. Conversely, if that tangential Hessian block vanishes throughout $\mathcal{X}^\circ$, take $P, P' \in \mathcal{X}^\circ$ differing only in the y -block. The segment between them lies in $\mathcal{X}^\circ$ by convexity of Δ_y° and the product structure, and $\phi(s) = \Pi_x \nabla_x E(P + s(P' - P))$ is $\mathcal{T}_x$ -valued with $\phi'(s)$ orthogonal to $\mathcal{T}_x$. Since $\mathcal{T}_x$ is closed, $\phi' \in \mathcal{T}_x \cap \mathcal{T}_x^\perp = \{0\}$, so ϕ is constant and $\neg(x \triangleright y)$. The middle condition is symmetric under $(x, u) \leftrightarrow (y, v)$ because $\nabla^2 E(Q)$ is a symmetric matrix at every Q (Schwarz's theorem; this is where (A1) is used). $\square$

Causality removes both directions. Write $C(x)$ for the Γ -connected component of x . Induction over an arbitrary schedule gives *confinement*: $Q_x^{(t)}$ is a function of the initial blocks $(Q_y^{(0)})_{y \in C(x)}$ and the words $(w_k)_{z_k \in C(x)}$ alone. Indeed, Lemma 6 makes the update of $x \in B_t$ a function of $\Pi_x \nabla_x E$ and, for a label node, of $w_{\pi(x)}$ and of no other word by (A3). Lemma 7, applied one block at a time inside $\mathcal{X}^\circ$, leaves that field a function of the blocks of x and its Γ -neighbours, all in $C(x)$. Head nodes need no separate treatment, since a route “through an H ” is a sequence of Γ -edges incident to that node. Confinement is a necessity statement (no path implies no influence), not a transmission statement: couplings can cancel or sit dead on the reachable set, which is what Proposition 5 exploits.

Proposition 8. Assume SE. Then C2 holds if and only if Γ has no edge between nodes at different positions.

Proof. Let $\pi(x) < \pi(y)$ and apply C2 to the one-node schedule $B_1 = \{x\}$ at $t = 1$: for every $Q^{(0)} \in \mathcal{X}^\circ$ the update of x ignores the y -block. Because C2 quantifies over all interior initialisations, this is an identical statement on $\mathcal{X}^\circ$, so $\neg(x \triangleright y)$ by Lemma 6 and then $\neg(y \triangleright x)$ by Lemma 7. This is the crux of the theorem: Schwarz symmetry converts “the later node's block does not feed the earlier node's update” into “the earlier node's block does not feed the later node's update”. The cross-position pair was arbitrary. Conversely, position-pure components make every $Q_x^{(t)}$ a function of

position- $\pi(x)$ initial blocks and of $w_{\pi(x)}$ only, by confinement, which is C2. $\square$

Proof of Theorem 1. Under SE and C2, Proposition 8 leaves Γ without a cross-position edge, so every component lies inside one position’s node set. Word entry is confined to label nodes by (A3)–(A4), and confinement then makes $Q_{z_i}^{(t)}$ a function of w_i and the position- i initial blocks alone, for every schedule, every t and every i . No label marginal ever depends on another word, which is $\neg X$. Contrapositively, a network-causal contextual scheme is not generated, in the sense of (A2), by any single C^2 word-local energy. $C2 \Rightarrow C1$ follows as well, word initialisations being particular interior ones; it holds vacuously, since nothing depends on any other word. Under a single energy the only way to be causal is to be trivial. The other half of the statement, that any two of the three properties hold together, is given by the three witnesses below. $\square$

For the masked scheme one can say more. One tangential Hessian entry is computed from the update at i and again from the update at $j < i$, whose masked message contains no Q_{z_i} at all. The two must agree, which forces $\Pi T^{(c)} \Pi = 0$ on every channel, and the centred label message of such a $T^{(c)}$ is a state-independent constant. A non-degenerate $T^{(c)}$ admits no energy, and a degenerate one carries no context. This is the formal content of Remark 2, and the reason the third witness below must give up SE.

The three witnesses. All three pairs are realisable, so the impossibility in Theorem 1 needs all three properties. $SE \wedge C2, \neg X$: take $E = \sum_i u_i(Q_i; w_i) + \sum_{i,c} \psi_{ic}(Q_{z_i}, Q_{h_{ic}})$, whose couplings are all within one position, so C2 holds by Proposition 8. $SE \wedge X, \neg C2$: the encoder model of Wu and Tu (2023), whose trilinear term $Q_{h_{jc}}(i) Q_j(b) Q_i(a) T_{b,a}^{(c)}$ makes a cross-position edge that Lemma 7 fires in both directions. $C2 \wedge X, \neg SE$: mask the head domains so that $H_i^{(c)}$ may point only to positions $j < i$, and keep only the message term weighted by $Q_{h_{ic}}(j)$. Every activation at position i is then a function of blocks at positions $\leq i$ for arbitrary interior states, X holds for generic $T^{(c)}$, and $\neg SE$ is forced by Theorem 1.

Back to the model. The directed chain of Section 2.3 gives up the single global energy and keeps causality and contextuality. It is not the third witness: that scheme deletes a term from a

	PTB	WikiText-2
Vocabulary (types)	10,000	33,278
Train tokens	929,589	2,088,628
Validation tokens	73,760	217,646
Test tokens	82,430	245,569
Validation blocks	1,152	3,400
Scored validation tokens	72,576	214,200
Test blocks	1,287	3,837
Scored test tokens	81,081	241,731
Unigram val. ppl.	688.82	964.82
slot 0 included	687.45	965.37
Bigram val. ppl.	445.24	692.50

Table 3: The two corpora as the loader sees them. Types include `<eos>` and are counted on the training split; no validation or test type is missing from it. Blocks are the non-overlapping $n = 64$ -token evaluation blocks; 63 tokens per block are scored (slot 0 is dropped for every model). The bigram is add-one smoothed towards the unigram.

symmetric field, whereas the chain never forms one (Remark 2). The energy is not simply dropped either. Each step conditional is generated by its own energy E_i , every update is exactly $\text{softmax}(-\nabla E_i / \lambda)$ with nothing deleted, and the global object is the valid directed joint $\prod_i p(z_i, h_i \mid z_{<i}, w_{1:i})$. Inference is variational filtering, so the belief at i approximates $p(Z_i \mid w_{1:i})$ and not the smoothing marginal $p(Z_i \mid w_{1:n})$. For a decoder, filtering is the right object.

D Datasets and Training Setup

Both corpora are word-level and pass through one reader, one tokenisation function and one evaluation protocol. All tuning is on Penn Treebank; WikiText-2 serves only for the transfer test of Section 4.3.7. Table 3 lists counts measured on the token streams the runs consume.

Penn Treebank. We use the standard word-level language-modelling version of the Penn Treebank (Marcus et al., 1993), with the preprocessing of Mikolov et al. (2012) as distributed: lower-cased, numbers replaced by N, rare words replaced by `<unk>`, punctuation removed, one sentence per line. The only type with an upper-case letter is N, which accounts for 32,481 of the 929,589 training tokens. Only 46 training types contain a digit, and all are compounds, decades or ordinals (*10-year*, *1920s*, *19th*). The only types with neither a letter nor a digit are #, \$, &, the apostrophe and two escaped asterisks. The reader appends `<eos>` to every line, so the sentence boundary is a token the model must

predict. There are 42,068 training sentences, 3,370 validation and 3,761 test sentences, and `<eos>` is 4.57% of the validation tokens. The vocabulary is built from the training split alone: 9,999 word types plus `<eos>`, the standard 10,000. No validation or test type is missing from it, so the reader’s `<unk>` fallback never fires and the unigram baseline needs no smoothing.

WikiText-2. We use the word-level release of Merity et al. (2017) (wikitext-2-v1, not the -raw- release, whose preprocessing differs). Its files are already mapped to a closed vocabulary, and `<unk>` accounts for 54,625 training tokens. They are read by the same function as PTB, with the same whitespace split, one `<eos>` per line and a vocabulary built from the training split: 33,278 types including `<eos>`, again with no validation or test type missing. The files keep the article structure. Of the 3,760 validation lines, 1,299 are blank and 620 are headings (= Heading =, = Sub =, and so on, up to five levels). We score blank lines and headings as they are, so that the token count matches the released files; a blank line contributes a bare `<eos>`. On the validation split blank lines are 0.60% of the tokens and heading lines 2.37%. Filtering the headings would change the denominator of the perplexity and make our numbers incomparable with published ones.

Batching and slot alignment. Each split is one `<eos>`-joined stream, cut into blocks of $n = 64$ tokens as in nanoGPT (Karpathy, 2022), and every model sees the same blocks in the same order. Training draws blocks at uniformly random offsets, so a block usually starts mid-sentence. The causal PT’s logits $[:, t]$ predicts w_t from $w_{<t}$ (Section 2.4), so it scores all n tokens of a block, including w_0 from ROOT alone. A causal transformer predicts only $w_1, \dots, w_{n-1}$. To score the same token set, the transformer wrapper runs the backbone on $w_{0:n-2}$ and shifts its logits right by one slot, filling slot 0 with zeros. Both models then drop slot 0 in training and in evaluation, so neither receives gradient from a slot the other cannot predict, and the wrapper’s loss asserts that the zero-filled slot is never scored. Both models are thus asked for $p(w_t \mid w_{<t})$ for $t = 1, \dots, n-1$: 63 scored tokens per block, 72,576 on PTB validation and 214,200 on WikiText-2 validation.

Evaluation. The evaluation loop walks fixed, non-overlapping blocks in corpus order over the

whole split, in batches of 16, and samples nothing. Trailing tokens that do not fill a block are dropped: 53, 32 and 62 tokens on the three PTB splits and 52, 46 and 1 on WikiText-2, at most 0.08% of a split. Perplexity is the exponential of the mean negative log-likelihood over all scored tokens of the split, not a mean of per-block perplexities. Training perplexity is measured the same way on a fixed 320-block slice of the training split.

Unigram baselines. The unigram reference of Section 4.1 is the maximum-likelihood unigram of the training split. It is scored through the same loop, so that it has the same 63 tokens per block as every model: 688.82 on PTB validation and 964.82 on WikiText-2 validation. Including slot 0 gives 687.45 and 965.37, a difference of at most 0.2%, in opposite directions on the two corpora. A model that does not clear the unigram by a wide margin has not learned to use context, whatever its loss curve looks like.

WikiText-2 is not a second tuning corpus. Every hyperparameter (learning rate and schedule, width, iteration count, temperature and damping) is selected on PTB and reused unchanged on WikiText-2; no grid is rerun there. This is what makes Section 4.3.7 a transfer test. Kuang et al. (2026) make no claim that their transfer method holds across corpora, so this test is not covered by their results. Two things change with the corpus: the vocabulary grows from 10,000 to 33,278 types, and the training set from 0.93M to 2.09M tokens. The word-label factor is $|\mathcal{V}| \times d$, so the parameter count changes even at fixed d , and WikiText-2 results are compared only with baselines trained on WikiText-2 through the same loop. The read-out bound of Proposition 4 also binds harder, with 3.3 times the vocabulary to place in an affine family of dimension d . It binds on a transformer’s output head in the same way at matched width (Section 3.3), but a weaker absolute number on WikiText-2 than on PTB partly reflects it.

Training protocol. Every model is trained by one loop, which reaches model-specific behaviour only through three methods: the loss on a scored token set, an L_2 regulariser on the arc scores (zero for a transformer) and a parameter count. PTB has under a million training tokens, so 15,000 steps of 16 blocks is 16.5 epochs, and the larger baselines reach their validation optimum long before the end (a 1.2M-parameter transformer at step 5,000,

against 15,000 for the causal PT). Every model is therefore scored at its own best-validation check-point, and test perplexity is measured on that check-point. Dropout is 0 for every model: Wu and Tu (2023) do not specify where it would enter the update equations, and any placement would add a map with no factor behind it (Section 2.6). The causal PT’s word unary b is set to the log unigram frequency of the training split and not trained. Runs used NVIDIA A40, TITAN RTX and L20 GPUs.

Model configurations. The baselines are nanoGPT decoders with four layers and four heads at widths d_{model} from 16 to 256; the looped transformer applies one such block four times with shared weights. Unless stated otherwise the causal PT uses $h = 2$ channels, $T = 3$ iterations, clip $\gamma = 3$, $\rho = 2$ predictive rounds (Appendix A.1), the mean-field readout, initialisation scale 0.02 and a learning rate of 2×10^{-2} . Following the step size of Wu and Tu’s Appendix B.1, the label update can be damped, $Q_{Z_i}^{(t)} = \alpha_Z Q_{Z_i}^* + (1 - \alpha_Z) Q_{Z_i}^{(t-1)}$, where $Q_{Z_i}^*$ is the undamped update and t indexes iterations; $\alpha_Z = 1$ means no damping. This differs from their message damping (their Appendix B.2). The scaling ladders, the gain and rank grids and the global-variable grid run undamped under the standardised temperature; the constants, depth and readout sweeps use the damped configuration ($\alpha_Z = 0.25$, original temperature, $d = r = 32$); the factored-label grid uses $\alpha_Z = 0.25$ and $h = 4$ at the original temperature; the switch ladder runs at $d_{\text{model}} = d = 64$.

E Implementation Validation

Two of the checks below, the brute-force comparison and the free-energy test, test the derivation; the rest test the software. All of them run the real code path on small tensors, in double precision wherever a precision claim is made.

Exact readout against brute force. On one slot the graph is a star centred on Z_t , hence a tree, so Equation (6) must equal the marginal of the slot joint computed by enumeration. We enumerate $p(W_t, Z_t, H_t^{(1..h)} \mid z_{<t})$ over every configuration, written from the raw parameters with loops and calling no method of the model, and marginalise to $p(W_t)$. The result agrees with the closed form to 6.7×10^{-16} in log-probability ($|\mathcal{V}| = 6$, $d = 4$, $h = 2$, $|\mathcal{D}_t| = 5$, float64). This checks the factorisation itself, not only its implementation. The

factored variant at $K = 2$ agrees with an enumeration over $(W_t, Z_t^{(1)}, Z_t^{(2)}, H_t^{(1..4)})$ to 10^{-10} . A second reference rebuilds the contracted arc scores by hand; it pins the assignment of channels to components, which the enumeration alone cannot see.

Global variable and readout scan. The same enumeration with a global variable folded in as one more leaf off Z_t agrees with Equation (21), and the logcumsumexp scan of Equation (19) agrees with slot-by-slot assembly for $\gamma \in \{0, 1, 2, 4, 9\}$; both are asserted at 10^{-12} .

Free energy, with a mutation test. The per-slot free energy $\mathcal{F}_t$ of Equation (12) must not increase along the inner loop, since each update is its centred gradient step. Over eleven block updates it shows 0 increases and a total decrease of 5.68×10^{-3} . This is the check that tests the update equations: a sign error in the head update would pass every shape, normalisation and causality check. With the head message negated, the same eleven updates give 11 increases, the largest 6.5×10^{-2} , and the free energy rises overall.

Causality. Changing the token at position $t+1$ leaves the logits at every slot $\leq t$ bitwise unchanged, under both readouts and all three head temperatures. Over the six combinations the maximum absolute difference is 0, checked by equality and not by a tolerance. The check runs on CPU with a fixed seed, because non-deterministic reductions on a GPU can change the last bit of correct code.

Frozen prefix in the gradient. The prefix is frozen in the forward pass but not detached in the backward pass (Section 2.3). A test on a detached tensor would pass trivially, so the prefix beliefs are kept as attached leaves. After backpropagating the loss at slot t , $\partial \mathcal{L}_t / \partial \bar{q}_j$ is exactly zero for $j \geq t$ and non-zero for $j < t$.

Tying. S appears once in the parameter list, and the same tensor object serves as the unary and as the output factor (Section 2.4). The test checks identity, not equality: two tensors initialised alike and updated by the same gradient would be equal but would form a different model.

Factored labels reduce to the flat model. With $K = 1$ the factored decoder must be the model of Section 2.3. A flat state dict loads into it with no missing or unexpected key, and the logits agree

to 6.9×10^{-18} across both readouts and both arc-score parameterisations in float64. The residual is floating-point reassociation from the extra component axis.

Normalisation and the worked example. Every posterior sums to one along its variable axis at every iteration (maximum deviation 1.2×10^{-7} in float32), the logits are finite, and a single batch can be overfitted to near-zero loss. The worked example of Appendix F matches an independently derived reference digit for digit.

All tables and figures are generated by one script from the run records, which store each run’s configuration, validation and diagnostic traces, device and code version.

F Worked Example

The example is small enough to check by hand.

Setting. $\mathcal{V} = \{the, cat, sat, mat\}$, two labels which we read as $\{N, V\}$, one channel, $\lambda_Z = \lambda_H = \lambda_W = 1$, a distance-insensitive arc score ($\gamma = 0$), one inner round per observed slot and $\rho = 2$ rounds in predictive mode. The factors are

$$S = \begin{pmatrix} 1.5 & -1.5 \\ 2 & -2 \\ -2 & 2 \\ 2 & -2 \end{pmatrix}, \quad T^{(0)} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix},$$

$$b = (1, 0, 0, 0), \quad r^{(0)} = (0, 1.5).$$

The prefix is *the cat sat*, and the model is asked for the word at slot 4.

Observed mode. Each slot initialises from its own unary, $Q_Z^{(0)} = \sigma(S_{w,\cdot})$, giving $(.953, .047)$, $(.982, .018)$ and $(.018, .982)$. One round of the head and label updates against the frozen prefix then gives the filtering marginals

$$\bar{q}_1 = (.818, .182), \quad \bar{q}_2 = (.957, .043),$$

$$\bar{q}_3 = (.006, .994).$$

The third word is labelled as the verb. Contracting these beliefs into arc scores by Equation (2) gives the cache that the next slot reads, $B_{1,\cdot}^{(0)} = (.365, .818)$, $B_{2,\cdot}^{(0)} = (.085, .957)$ and $B_{3,\cdot}^{(0)} = (1.987, .006)$. Nothing else of the prefix survives.

Predictive mode. The word variable is unclamped, so it initialises from its own unary factor, $Q_W^{(0)} = \sigma(b) = (.475, .175, .175, .175)$, which gives the query $\tilde{s} = (1.063, -1.063)$ and $Q_Z^{(0)} = (.893, .107)$. The query is the [MASK] embedding, produced by the same initialisation rule as every other variable. Two rounds of the inner loop over the head domain $\{\text{ROOT}, 1, 2, 3\}$ give

$$F^{(1)} = (.160, .413, .178, 1.776)$$

$$Q_{H^{(1)}} = (.120, .154, .122, .604)$$

$$Q_Z^{(1)} = (.951, .049)$$

$$F^{(2)} = (.074, .387, .128, 1.890)$$

$$Q_{H^{(2)}} = (.104, .143, .110, .642)$$

$$Q_Z^{(2)} = (.956, .044)$$

The slot attends mostly to position 3, the verb, which is the head a noun object should choose. The word update is the output layer. It gives the logits $(2.368, 1.824, -1.824, 1.824)$ and

$$\hat{p}(W_4) = (.460, .267, .007, .267)$$

over (the, cat, sat, mat) . The verb is almost excluded, and *mat* and *cat* tie because nothing in this two-label toy separates two nouns.

Observing the word. Clamping $W_4 = mat$ and running the same slot in observed mode sharpens the label belief (entropy $0.180 \rightarrow 0.040$ nats) and raises the attention on the verb from .642 to .663. As the construction predicts, observing the word makes the slot more certain of both its label and its head.

The exact readout on the same slot. With the same parameters, Equation (6) gives $\hat{p}(W_4) = (.365, .215, .206, .215)$. This is flatter, and it no longer almost excludes the verb. The mean-field readout commits to one soft label and reads the vocabulary off it, while the exact readout mixes over labels, so a word compatible with an unlikely label keeps more mass. Section 4.3.6 compares the two readouts after training.

G What the Latent Variables Learn

The label and head variables are meant to carry syntactic categories and dependency attachments. We probe trained models to see what they carry.

Words that each label explains. The word–label factor is the only place where a word touches the graph, so the words a label explains are those with the largest $S_{w,a}$. Raw scores favour words with large rows, so we rank by the contrastive score $S_{w,a} - \max_{a' \neq a} S_{w,a'}$. Of $d = 32$ labels, several are recognisably syntactic:

label	top words by contrastive score
$z=6$	<i>reconsider condemn leave lend hire notify restore</i>
$z=30$	<i>prefers manages chooses earns requires bears helps</i>
$z=13$	<i>to affects between from until thus indeed besides</i>
$z=19$	<i>controlled punishable bolstered funded headed owned</i>
$z=20$	<i>henderson cook jr. fisher sr. robertson generale</i>
$z=26$	<i>pence marks miles equaling tons</i>
$z=22$	<i>speculation deficits view sums notion evidence</i>

In order, these are bare infinitives, third-person singular present verbs, prepositions and subordinators, past participles, person names, units of measure and abstract nouns. No supervision produced these categories; they come from maximising the likelihood through a 32-value bottleneck. About a third of the labels resist any reading we can offer, as [Wu and Tu \(2023\)](#) report for the encoder.

Where the head posterior looks. The head variable is the model’s dependency attachment, and here the configuration matters most. We probed two trained models: the damped configuration ($d = 32$, original temperature, damped) and the standardised configuration ($d = 96$, $r = 12$, standardised temperature, undamped; Section 4.3.3).

	damped	standardised
ROOT mass	0.486	0.277
mean attachment distance	1.05	11.06
mass at distance 1	0.976	0.327
mass beyond distance 3	0.001	0.497

The damped model has not learned attachment. It puts 0.976 of its positional mass at distance one and almost none beyond three: a bigram attachment with large root mass. The standardised model has a mean attachment distance of 11.1, against 17.0 for uniform attention over the same prefixes. It is 5.2 times uniform at distance one, yet still places half its mass beyond distance three (Figure 4). This is a preference for nearby heads, not a collapse onto the previous word.

The two models give opposite answers to the same probe; what separates them is the head temperature and the arc-score rank. An interpretability result for this model class says little without the parameterisation that produced it.

Predictive against observed beliefs. We measure the KL divergence, in nats, from a slot’s predictive label belief to its observed one. We expect it

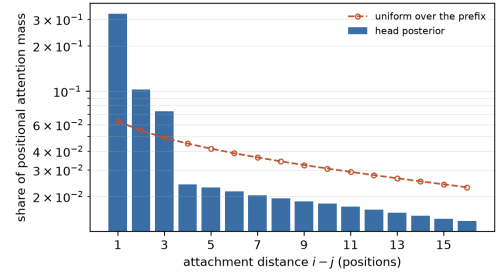


Figure 4: Head-posterior attachment distance in the standardised configuration ($d = 96$, $r = 12$) against uniform attention over the same prefixes. The posterior is 5.2 times uniform at distance one, yet half its mass lies beyond distance three; mean distance 11.1 against 17.0. Distances 1–3 have their own arc scores ($\gamma = 3$); longer ones share one.

to be small on predictable tokens and to grow with the surprisal of the observed word. Over 4,032 slots per model the expected direction appears in both models, and its strength separates them as the attachment probe does. In the standardised configuration the correlation is $r = 0.47$, with a mean divergence of 4.07 on the most predictable quartile of tokens against 6.56 on the most surprising. In the damped configuration it is $r = 0.16$, with 1.98 against 2.58. The expectation concerns the direction only, so we read nothing into the magnitudes beyond this: the parameterisation decides how much of the predicted structure is there to find.

Width. The label lexicon above comes from a model with $d = 32$, where each label has enough words to be readable. At $d = 96$ the same probe finds fewer clean categories and more fragments. Adverbs (*hastily, mildly, extremely, widely*), gerunds (*completing, entering, joining, contemplating*), units (*bottle, par, pound, capita, bushel*) and time nouns (*century, decade, phase, 1970s*) survive, but most labels resist reading. Interpretability and capacity pull in opposite directions along the width axis, and we found no width at which both are best.

H Additional Results

Cells are means $\pm$ sample standard deviation over the stated number of seeds. A seed that finishes above 95 % of the unigram perplexity counts as collapsed and is not averaged (except in Table 6): *n/m collapsed* means *n* of *m* seeds collapsed. An italic range replaces the mean when no seed collapsed but the seeds disagree (largest at least $1.5\times$ the smallest, or a standard deviation above 15 % of

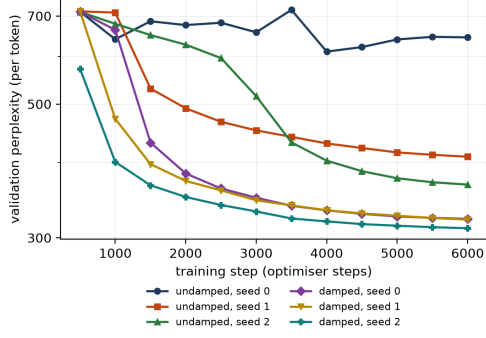


Figure 5: One undamped configuration at the original temperature, three seeds each without and with damping ($\alpha_Z = 0.25$). Undamped runs separate by more than 200 perplexity and do not reconverge; damped runs are indistinguishable after 3,000 steps. Curves are validation traces; the text quotes best-validation checkpoints, so a curve may end above its quoted value.

the mean).

H.1 Reproducibility

The undamped configuration at the original temperature ($d = 16$, 6,000 steps) ends at 430.0 on an NVIDIA A40 and at 645.0 on a TITAN RTX (Section 4.3.1). A rerun on an A40 reproduces the whole twelve-point validation trace to every printed digit; the TITAN RTX trace begins 711.6, 640.3, 686.0. The first evaluations already differ (711.70 against 711.61), so the runs diverge within 500 steps; the gap then grows to 215 perplexity. Nothing differs but the order of floating-point reductions: on one device, two versions of the implementation and two training drivers, in three combinations, agree to every printed digit.

Seeds act the same way. Three seeds on one device give 610.9, 408.8 and 367.5 (462.4 ± 130.2). The same 3 seeds damped with $\alpha_Z = 0.25$ give 318.4 ± 6.3 , a standard deviation of 2.0% of the mean, 21 times tighter (Figure 5). In this regime single-seed comparisons are not informative. A global variable moves an undamped model at $d = 24$ from 321.4 to 297.7 with one seed each, a difference of 23.7 against a seed standard deviation of 130.2. A single-seed comparison of damping gives 641.5 against 248.3; damping improves reproducibility at least as much as the mean. The damped configuration at $d = 32$ is reproducible: 253.0 ± 6.0 over three seeds, against 250.7 ± 2.3 from an independent set of runs.

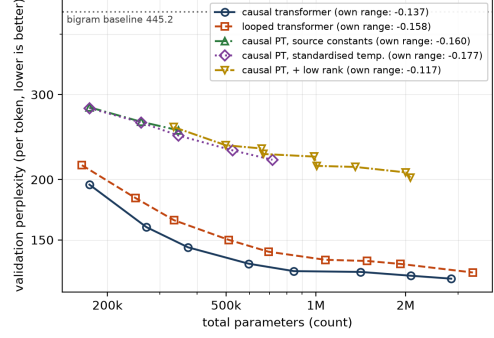


Figure 6: As Figure 3, against total parameters. With tied embeddings almost all of the causal PT’s parameters are in S .

H.2 Scaling

The transformer gives 129.2 at $d_{\text{model}} = 64$, 124.7 at 160 and 125.3 at 256, while its training perplexity falls from 68.5 to 10.2: on 929,589 training tokens it is limited by data. At matched non-embedding budget the looped transformer is better at small sizes (164.7 at 12,768 parameters against 195.2 at 13,152) and worse at large ones (135.7 at 198,528 against 129.2 at 200,064). At matched total parameters the standard transformer wins everywhere (Figure 6).

A model’s frontier is its best-of-two-learning-rates point at each size. Fitted over each curve’s own range, the exponents are -0.086 (transformer), -0.087 (looped), -0.090 (full-rank causal PT, standardised temperature) and -0.058 (low-rank causal PT). The first three look alike only because the baselines extend to 1.2×10^6 parameters, where flat points pull their slopes down; from the same runs, the looped transformer gives -0.042 over the transformer’s range and -0.130 over the causal PT’s. The full-rank causal PT is absent from Table 4 because the looped frontier has only two points on the range it shares with both baselines. In the pairwise fits of Table 5 the ordering is the same in all five pairs, and the leave-one-out ranges are disjoint in every pair with enough points to compute them.

Causal PT entries are seed means, and the best seed can differ from the mean by more than the effects we discuss: at $d = 192$, $r = 24$ the best of 3 seeds gives 197.3 and the mean is 206.9 ± 9.2 .

H.3 Width, temperature and rank

Width. Undamped at the original temperature, $d = 32$, 96 and 128 end at 700.7, 698.1 and 701.4 (Figure 7). The best damping step is at the edge

	n	exp.	R^2	ppl across it
transformer	4	-0.175	0.95	189 $\rightarrow$ 124
looped	3	-0.068	0.96	157 $\rightarrow$ 133
PT low rank	8	-0.045	0.93	226 $\rightarrow$ 203

Table 4: Scaling exponents fitted on the one range all three curves share, 13k–148k non-embedding parameters; n is the number of frontier points per fit.

of the grid at five of seven widths, and we did not sweep below $\alpha_Z = 0.125$. With the damped constants tuned at $d = 32$, the model becomes unpredictable at larger widths, not gradually worse: 253.0 ± 6.0 at $d = 32$, 287.4 ± 52.5 at 48 and 470.0 ± 160.7 at 64. The growing standard deviation matters more than the mean. Under the standardised temperature, undamped, the full ladder reaches 229.8 ± 8.5 at $d = 48$ and 219.8 ± 4.2 at $d = 64$ (Table 6). Figure 9 separates two failures that end on the same unigram perplexity: at $d = 96$ the beliefs never leave the uniform point, while at $d = 32$ they sharpen to 5.5, hold for about 2,000 steps and collapse, the feedback loop of Section 3.2.

Gain. We swept the gain over $\{1, 1.5, 2, 3, 4, 6\}$ at four widths (Table 7). The argmin does not depend on width because, after standardisation, the attention entropy depends on the gain and the size of the head domain but not on d . At $d \geq 96$ the working range shrinks to gain 1, the width ceiling of Section 4.3.3. Above the range the failure is reproducible: at $d = 32$, gain 3 collapses all three seeds (700.5 ± 0.7).

Learning rate. Table 8 is the diagnostic used for maximal update parameterisation (Yang et al., 2021), which Kuang et al. (2026) apply to the encoder. Over an eightfold range of width the argmin moves by one step of the $2 \times$ grid in both arms: from 2×10^{-2} to 10^{-2} under the original temperature and from 4×10^{-2} to 2×10^{-2} under the standardised one. With one seed per cell, the argmin under the original temperature is within seed noise.

Rank. The arc scores factor as $T^{(c)} = U^{(c)}V^{(c)\top}$ with $U^{(c)}, V^{(c)} \in \mathbb{R}^{d \times r}$. Wu and Tu (2023) set $r = \min(64, d)$ in their Table 2, which for $d \leq 64$ is $r = d$, the most expensive choice. At $d = 96$, halving the rank to $r = 48$ improves perplexity and cuts the seed spread eightfold (Table 9); $r = 12$ gives 257.8 ± 4.5 with eight times fewer non-embedding parameters and still beats the best

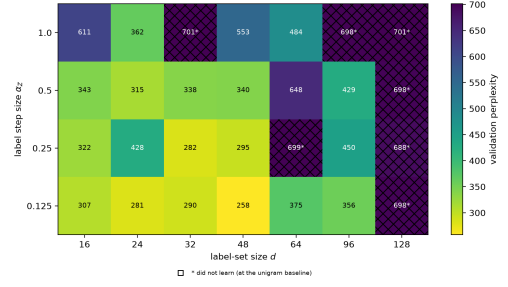


Figure 7: Label-set size against damping step α_Z on PTB at the original temperature $\lambda_H = 1/d$, 6,000 steps; hatched cells finished on the unigram, grey cells were not run.

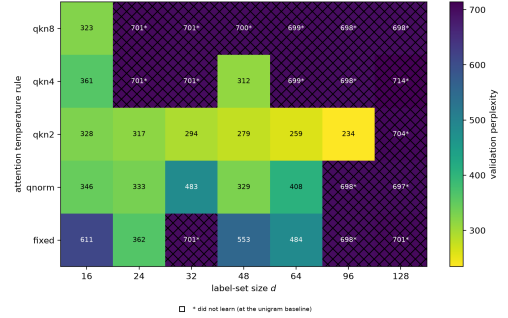


Figure 8: Label-set size against the head-temperature rule on PTB, undamped, 6,000 steps. ‘fixed’ is the original temperature, ‘qnorm’ the belief-norm temperature and ‘qkng’ the standardised temperature at gain g ; hatched cells finished on the unigram.

$d = 32$ configuration of comparable size. The arc score enters inference only as $B_{j,a}^{(c)} = \mathbb{E}_{\tilde{q}_j}[T_{a,\cdot}^{(c)}]$, so only its rank on the affine hull of the reachable beliefs is used, and these beliefs are measurably concentrated. On the full ladder, low rank sits below $r = d$ (223.1 ± 5.6 at 18,624 non-embedding parameters against 372 ± 277 for $r = 96$ at 147,648, a mean that includes one collapsed seed). It trains stably through $d = 256$, $r = 32$, with seed standard deviations from 1.5 to 10.4 over the 11 configurations that trained (Table 1), but one seed in three still collapses at $d = 256$, $r = 64$, the widest and highest-rank cell.

H.4 Global variables and initialisation

An initialisation has a treatment (whether B' shares the scale and L_2 coefficient of the other factors) and a draw distribution (Gaussian unless stated). A symmetry can stall the global variable. If the rows of B' are close relative to their length, the variable’s posterior is uniform, the message is the row mean, and every row gets the same gradient $1/m$, so nothing separates them, and a shared L_2

pair	range	n	exponent	leave-one-out	n	exponent	leave-one-out
transformer / looped	13k–790k	6	−0.115	−0.148 to −0.078	6	−0.042	−0.045 to −0.034
looped / causal PT	4.1k–66k	4	−0.130	−0.145 to −0.110	5	−0.090	−0.093 to −0.090
looped / PT low rank	4.2k–148k	5	−0.107	−0.130 to −0.088	10	−0.058	−0.062 to −0.046
transformer / causal PT	13k–66k	3	−0.223	n too small	3	−0.084	n too small
transformer / PT low rank	13k–148k	4	−0.175	−0.223 to −0.126	8	−0.045	−0.049 to −0.042

Table 5: Pairwise exponents, each pair fitted on the range both curves share: number of frontier points, exponent, and its range when any one point is left out. Where the two leave-one-out ranges in a row do not overlap, no single point decides the ordering. Two rows rest on three points a side, the minimum for a fit and too few to leave one out.

Model	Config	Emb.	Non-emb.	Val ppl	Test ppl
Unigram	—	—	—	688.8	—
Causal PT	$d = 16$	170,000	4,128	281.1 ± 8.5	257.2 ± 7.1
Transformer	$d_{\text{model}} = 16$	161,024	13,152	195.2	181.0
Looped	$d_{\text{model}} = 16$	161,024	3,312	214.2	196.5
Looped	$d_{\text{model}} = 24$	241,536	7,272	183.1	168.4
Causal PT	$d = 24$	250,000	9,264	262.5 ± 13.6	240.4 ± 12.6
Transformer	$d_{\text{model}} = 16$	161,024	13,152	195.2	181.0
Transformer	$d_{\text{model}} = 24$	241,536	28,944	159.4	147.2
Looped	$d_{\text{model}} = 24$	241,536	7,272	183.1	168.4
Looped	$d_{\text{model}} = 32$	322,048	12,768	164.7	150.1
Causal PT	$d = 32$	330,000	16,448	246.7 ± 2.7	224.7 ± 3.5
Transformer	$d_{\text{model}} = 24$	241,536	28,944	159.4	147.2
Transformer	$d_{\text{model}} = 32$	322,048	50,880	144.8	133.9
Looped	$d_{\text{model}} = 32$	322,048	12,768	164.7	150.1
Looped	$d_{\text{model}} = 48$	483,072	28,368	150.0	137.2
Causal PT	$d = 48$	490,000	36,960	229.8 ± 8.5	210.3 ± 7.5
Transformer	$d_{\text{model}} = 32$	322,048	50,880	144.8	133.9
Transformer	$d_{\text{model}} = 48$	483,072	113,184	133.9	122.3
Looped	$d_{\text{model}} = 48$	483,072	28,368	150.0	137.2
Looped	$d_{\text{model}} = 64$	644,096	50,112	141.8	129.7
Causal PT	$d = 64$	650,000	65,664	219.8 ± 4.2	201.3 ± 3.4
Transformer	$d_{\text{model}} = 48$	483,072	113,184	133.9	122.3
Transformer	$d_{\text{model}} = 64$	644,096	200,064	129.2	118.2
Looped	$d_{\text{model}} = 64$	644,096	50,112	141.8	129.7
Looped	$d_{\text{model}} = 96$	966,144	112,032	136.4	125.9
Causal PT	$d = 96$	970,000	147,648	372.4 ± 277.0	340.4 ± 251.2
Transformer	$d_{\text{model}} = 64$	644,096	200,064	129.2	118.2
Transformer	$d_{\text{model}} = 96$	966,144	447,552	128.8	118.7
Looped	$d_{\text{model}} = 96$	966,144	112,032	136.4	125.9
Looped	$d_{\text{model}} = 128$	1,288,192	198,528	135.7	124.4
Causal PT	$d = 128$	1,290,000	262,400	508.9 ± 186.9	465.6 ± 171.6
Transformer	$d_{\text{model}} = 96$	966,144	447,552	128.8	118.7
Transformer	$d_{\text{model}} = 128$	1,288,192	793,344	126.5	118.1
Looped	$d_{\text{model}} = 128$	1,288,192	198,528	135.7	124.4
Looped	$d_{\text{model}} = 160$	1,610,240	309,600	133.8	125.3

Table 6: The full-rank ladder ($r = d$) under the standardised temperature, each causal PT row (three seeds) followed by the nearest measured baselines (better of two learning rates). Embedding and non-embedding parameters are listed separately. Unlike the other tables, rows here average collapsed seeds in.

penalty shrinks whatever spread could.

Original temperature. At $d = 32$ the model without a global variable collapses (700.7), while several configurations with one reach 323 to 380 (6,000 steps), as if the extra message into Z_t broke the feedback loop. With one seed per cell (Table 10), Gaussian, uniform and semi-orthogonal draws give 700.7, 375.0 and 386.3 at $d = 32$. Against a seed standard deviation of 130.2 we draw no conclusion from this grid.

Standardised temperature. With three seeds, undamped (Table 12), the global variable has no effect at $d = 32$: 243.2 ± 5.3 without it, 242.5 to 254.1 with it across three scales and both L_2

treatments; the collapse it rescued at the original temperature is gone. At $d = 64$, the edge of the stable region, the model without it is unstable: one seed in three collapses (216.8, 215.1 and 704.1). With it, four of six treatments train on every seed (221 to 234). The two that do not are the smallest initialisation outside the shared L_2 (224.0, 616.5, 639.1) and the middle one inside it (215.0, 235.2, 667.0). The largest, 0.5, is stable under both treatments; this is the one regularity here, and the opposite of what the symmetry account predicts. The global variable changes which configurations are stable; it does not make the model stable. Otherwise, Gaussian scales spanning $25\times$, with or without the shared L_2 , change little else.

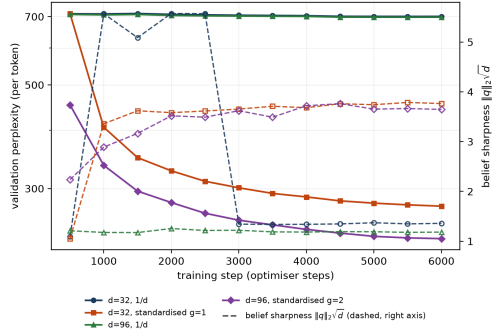


Figure 9: Within-run mechanism, undamped. Solid: validation perplexity (left axis). Dashed: belief sharpness $\|Q_Z\|_2 \sqrt{d}$ (right axis), 1 at the uniform belief and $\sqrt{d}$ at a one-hot. At the original temperature the $d = 96$ run never leaves the uniform point and the $d = 32$ run sharpens to 5.5, holds, then collapses. Under the standardised temperature sharpness rises to a moderate value and stays, and perplexity falls throughout.

Entropy and capacity. The entropy $H(Q_g)$ of the global variable’s posterior is at least 0.93 of its maximum in every cell, even where the variable’s message is 1.19 to 1.52 times the word unary (outside the shared L_2). The variable thus sends a near-constant label prior, not the feed-forward-like operator it was proposed as, and perplexity alone cannot show this. Capacity agrees: $m = 16, 64$ and 256 give 224.2 ± 5.1 , 222.2 ± 4.8 and 222.8 ± 4.7 , so sixteen times more values are worth less than one seed standard deviation.

Draw. At $d = 64$ (Table 11), the semi-orthogonal draw’s worst seed beats the Gaussian mean, and the uniform draw collapses on one seed in three. For a model whose main practical problem is seed variance, this matters more than the difference in means. The symmetry account suggests that a shared L_2 should stall a small Gaussian draw and spare an orthogonal one. Instead the small Gaussian draw fails outside the shared L_2 , and the orthogonal draw’s advantage is its low variance. Maximal update parameterisation (Yang et al., 2021) assumes a Gaussian draw, so its transfer results may not be neutral to this choice.

H.5 Switch ladder

S1. Value tying, which the factor graph forces because both roles are the same message through the same arc factor, costs -0.6 . Layer normalisation costs -0.7 once each switch is tuned; the 18.5 it seemed to cost at a single learning rate came from the rate. With the posterior readout of Section 2.4,

d	gain	n	val ppl	collapsed
32	1.0	3	282.2 ± 6.6	0
32	1.5	3	302.7 ± 26.6	0
32	2.0	3	290.5 ± 8.4	0
32	3.0	3	700.5 ± 0.7	3
32	4.0	3	570.9 ± 225.1	2
32	6.0	3	616.0 ± 147.1	2
64	1.0	3	254.0 ± 1.9	0
64	1.5	3	262.3 ± 3.6	0
64	2.0	3	260.3 ± 0.8	0
64	3.0	3	415.3 ± 245.2	1
64	4.0	3	631.9 ± 116.0	2
64	6.0	3	673.8 ± 49.0	2
96	1.0	3	275.6 ± 27.1	0
96	1.5	3	548.0 ± 276.0	2
96	2.0	3	418.4 ± 260.6	1
96	3.0	3	702.6 ± 7.4	3
96	4.0	3	582.1 ± 201.3	2
96	6.0	3	698.6 ± 0.5	3
128	1.0	3	433.6 ± 205.6	0
128	1.5	3	601.7 ± 186.4	2
128	2.0	3	709.7 ± 5.9	3
128	3.0	3	690.6 ± 13.1	3
128	4.0	3	704.6 ± 11.1	3
128	6.0	3	698.1 ± 0.4	3

Table 7: Validation perplexity by label-set size and standardisation gain; three seeds, undamped, 6,000 steps; ‘collapsed’ counts seeds on the unigram. The argmin is 1 at every width; the working range is $[1, 2]$ at $d \in \{32, 64\}$, and at $d \in \{96, 128\}$ only gain 1 avoids collapse.

d	temperature	0.005	0.01	0.02	0.04	0.08	argmin
16	source	697	641	430	662	708	0.02
16	standardised	697	701	334	300	305	0.04
32	source	696	502	674	716	710	0.01
32	standardised	696	350	294	288	714	0.04
64	source	696	359	664	700	711	0.01
64	standardised	696	658	258	710	714	0.02
128	source	694	499	607	716	709	0.01
128	standardised	694	442	416	713	721	0.02

Table 8: Validation perplexity against learning rate; one seed, undamped, 6,000 steps. ‘source’ is the original temperature $\lambda_H = 1/d$, ‘standardised’ the standardised one at gain 2. In both arms the argmin moves by one grid step over an eightfold range of width.

at matched width and with tied embeddings, a transformer does not train: it ends at 686.8 (training perplexity 736.6), and no rate from 10^{-3} to 10^{-2} does better. The evaluation swap of Section 4.3.6 shows weights trained for one readout failing under the other; here a readout moved into a model built for the other cannot be trained. Together the two show that the mixture readout is not a drop-in replacement for a linear head. This switch keeps a free projection into label space (see Caveats), so it is not the causal PT’s exact readout.

The simplex state’s $+564.3$ is not the cost of the simplex alone. A transformer whose state is softmaxed after every block has a state norm in $[d^{-1/2}, 1]$ while its attention still divides by $\sqrt{d_{\text{head}}}$,

d	r	non-emb.	n	val ppl
32	4	2,112	3	308.4 ± 17.2
32	8	4,160	3	295.8 ± 1.1
32	16	8,256	3	294.3 ± 3.6
32	32	16,448	3	282.2 ± 6.6
96	12	18,624	3	257.8 ± 4.5
96	24	37,056	3	256.2 ± 3.8
96	48	73,920	3	242.6 ± 3.3
96	96	147,648	3	275.6 ± 27.1

Table 9: Validation perplexity against the rank r of the arc scores; three seeds, standardised temperature, undamped, 6,000 steps.

d	draw	head	val ppl	test ppl
16	normal	no	430.0	393.1
16	orthogonal	no	346.1	314.0
16	uniform	no	424.4	386.3
32	normal	no	700.7	641.9
32	orthogonal	no	386.3	352.3
32	uniform	no	375.0	341.4
32	normal	yes	465.0	425.1
32	orthogonal	yes	344.9	310.0
32	uniform	yes	508.8	454.2

Table 10: Initialisation draw at the original temperature, undamped, one seed per cell, 6,000 steps. The seed sd of this configuration is 130.2 (Section 4.3.1) and the largest difference here is 326, so no conclusion is drawn; Table 12 repeats the question at three seeds.

the miscalibration of Section 3.2 seen from the transformer side. The ladder has no temperature switch, so this interaction lands on the state. The causal PT, with a simplex state and a temperature suited to it, reaches 206.1. In Figure 10 the interaction shows as a spike that the readout flip partly undoes.

S2. Restoring nine of the ten transformer properties makes the all-PT model worse, by between 21.5 (weight sharing) and 392.8 perplexity (an unconstrained state, which ends on the unigram). Value tying is the one neutral change (+0.7). The parts of the causal PT therefore depend on each other, which S1 cannot show: there, six of the same properties are free or nearly so.

Caveats. The ladder has one seed per cell (128 cells), so we draw conclusions only from effects larger than the seed spread measured in the same regime. The all-PT anchor is best at the top of its learning-rate grid: four rates gave 687.2, 395.5, 354.5 and 296.7, and a fifth, 10^{-1} , gave 277.3. The improvements (-292 , -41 , -58 , -19) have not turned over, and four of the ten S2 switches also sit at the new boundary. The S2 magnitudes are

draw	val ppl	seeds
semi-orthogonal	219.3 ± 0.6	218.8, 219.1, 219.9
Gaussian	225.8 ± 10.2	214.1, 229.9, 233.3
uniform	<i>1/3 collapsed</i>	225.2, 272.9, 707.1

Table 11: Initialisation draw of B' at $d = 64$, $m = 64$, scale 0.02, shared L_2 ; three seeds, with each seed’s value.

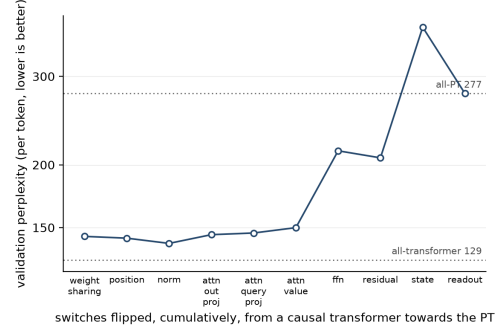


Figure 10: Cumulative switch ladder: each step flips one more difference from the causal transformer towards the causal PT and keeps the earlier ones. Each point is the best of the learning rates run for that step; dotted lines are the tuned endpoints. The last step is the all-PT model itself. Values in Table 13.

therefore lower bounds, though their signs are not in doubt. Every S1 optimum is interior, so the per-switch attributions rest on S1.

The all-PT model approximates the causal PT in four respects: its mixture readout keeps a free projection into label space where the causal PT reads the label posterior directly; relative positions enter as a scalar bias per head and distance bucket, not as a full arc-score matrix per bucket; heads are concatenated, not summed over channels; and there is no switch for the damping step or the head temperature. The ladder runs at $d_{\text{model}} = d = 64$ with four layers (iterations) and four heads (channels).

H.6 Constants and inference depth

All three sweeps use one seed per cell.

L_2 on the arc scores. Coefficients of 0 , 5×10^{-5} , 5×10^{-4} , 5×10^{-3} and 5×10^{-2} give 273.3, 255.6, 248.3, 249.8 and 697.0. At 5×10^{-2} the arc message is 2 % of the word unary and removing the prefix changes the prediction by a KL below 10^{-8} : the arc factors have been regularised away. Wu and Tu’s value is at or near the optimum.

Relative-position clip. $\gamma = 1$ and $\gamma = 3$ give 249.7 and 248.3; $\gamma = 0$ and $\gamma = 7$ give 619.4 and 605.1, and fail in opposite ways. At $\gamma = 0$

d	m	init sd	in L_2	dist.	n	val ppl	$H(Q_g)/\max$
32	0	—	yes	normal	3	243.2 $\pm$ 5.3	0.000
32	64	0.02	no	normal	3	249.3 $\pm$ 1.9	0.992
32	64	0.02	yes	normal	3	242.5 $\pm$ 5.4	1.000
32	64	0.1	no	normal	3	254.1 $\pm$ 4.4	0.976
32	64	0.1	yes	normal	3	246.6 $\pm$ 9.8	1.000
32	64	0.5	no	normal	3	248.4 $\pm$ 3.5	0.942
32	64	0.5	yes	normal	3	250.3 $\pm$ 9.9	0.973
64	0	—	yes	normal	3	<i>1/3 collapsed</i>	0.000
64	16	0.1	no	normal	3	224.2 $\pm$ 5.1	0.977
64	64	0.02	no	normal	3	<i>224–639</i>	0.993
64	64	0.02	yes	normal	3	225.8 $\pm$ 10.2	1.000
64	64	0.02	yes	orthogonal	3	219.3 $\pm$ 0.6	1.000
64	64	0.02	yes	uniform	3	<i>1/3 collapsed</i>	1.000
64	64	0.1	no	normal	3	222.2 $\pm$ 4.8	0.974
64	64	0.1	yes	normal	3	<i>215–667</i>	1.000
64	64	0.5	no	normal	3	221.4 $\pm$ 5.4	0.929
64	64	0.5	yes	normal	3	228.0 $\pm$ 11.6	1.000
64	256	0.1	no	normal	3	222.8 $\pm$ 4.7	0.989

Table 12: The global variable in the standardised configuration of Section 4.3.3: number of values m , initialisation sd, whether B' shares the L_2 of the other factors, draw, three seeds. $H(Q_g)/\max$ is the entropy of the global variable’s posterior relative to its maximum; 1.000 means uniform.

switch	S1 val	Δ	S2 val	Δ
attn_out_proj	130.8	+1.5	380.0	-102.7
attn_query_proj	131.1	+1.9	302.0	-24.7
attn_value	128.7	-0.6	276.6	+0.7
ffn	144.4	+15.1	530.7	-253.4
norm	128.6	-0.7	501.9	-224.6
position	128.3	-1.0	313.3	-36.0
readout	686.8	+557.6	375.2	-97.9
residual	136.7	+7.5	356.4	-79.1
state	693.5	+564.3	670.2	-392.8
weight_sharing	141.8	+12.5	298.8	-21.5

Table 13: Cost of each difference, measured in two directions. S1: the transformer with that one property replaced by the causal PT’s. S2: the all-PT model with that one property restored to the transformer’s. Δ is measured from the anchor of its own family (the tuned transformer for S1; the all-PT model, 277.3, for S2) and signed so that a positive value means the causal PT’s version is worse. Best of four learning rates per switch, one seed per cell. Switch names follow Section 3.4.

the model cannot see distance, and the KL from removing the prefix falls to 0.037. At $\gamma = 7$ the arc message reaches eight times the word unary and the feedback loop of Section 3.2 runs away.

Depth. Wu and Tu fix $T = 3$. In the damped configuration at $d = 32$ (seed standard deviation 6.3), $T = 1, 2, 3, 5, 8$ give 241.3, 246.5, 248.3, 297.0 and 275.3, and $\rho = 1, 2, 4$ predictive rounds (Appendix A.1) give 242.1, 248.3 and 284.9; the middle value is the shared base run ($T = 3, \rho = 2$). One seed cannot tell whether the lower value at $T = 8$ than at $T = 5$ is real. The extra iterations may be harmful, since each re-injects the same arc factors and Section 4.3.3 shows the feedback loop running away, or they may only make optimisa-

arm	config	non-emb.	n	val ppl	test ppl
fixed	$d = 32$	16,448	3	343.1 $\pm$ 24.7	321.5 $\pm$ 22.0
fixed	$d = 64$	65,664	3	334.1 $\pm$ 27.6	312.8 $\pm$ 25.7
gpt	$d_{\text{model}} = 32$	50,880	2	179.7 $\pm$ 4.3	167.9 $\pm$ 2.9
gpt	$d_{\text{model}} = 64$	200,064	2	150.9 $\pm$ 1.7	142.3 $\pm$ 2.1
gpt	$d_{\text{model}} = 128$	793,344	2	148.5 $\pm$ 5.4	141.6 $\pm$ 5.6
looped	$d_{\text{model}} = 32$	12,768	2	203.0 $\pm$ 4.9	188.6 $\pm$ 5.4
looped	$d_{\text{model}} = 64$	50,112	2	167.9 $\pm$ 0.7	159.1 $\pm$ 1.0
looped	$d_{\text{model}} = 128$	198,528	2	157.2 $\pm$ 1.3	149.6 $\pm$ 0.7
rule	$d = 32$	16,448	3	308.1 $\pm$ 2.6	288.6 $\pm$ 2.2
rule	$d = 64$	65,664	3	279.1 $\pm$ 9.0	260.3 $\pm$ 9.1

Table 14: WikiText-2 with every hyperparameter carried over from PTB. ‘fixed’ and ‘rule’: the causal PT under the original temperature (damped) and the standardised temperature (undamped), three seeds; ‘gpt’ and ‘looped’: causal and looped transformer (two seeds). Unigram 964.8, bigram 692.5.

tion harder. Either way, these results do not show whether computation shared across depth helps at larger scale.

H.7 Readouts

Swapping the readout at evaluation isolates the approximation error; training under each readout measures the end-to-end effect. The swap is meaningful only at $\lambda_W = 1$ (Appendix A.3). Over the 11 configurations with three uncollapsed seeds, the swap takes seed-mean perplexity from 223.1 to 1621 at the mildest ($d = 96, r = 12$) and from 225.9 to 4146 at the worst ($d = 64, r = 16$). The exact logits are not degenerate: they discriminate less between words at a fixed context (standard deviation 1.63 against 2.64 for the mean-field readout) and more between contexts for a fixed word (2.73 against 2.42). Trained under each readout (one seed each), the exact one is worse by factors of 1.41 ($d = 32$) and 1.30 ($d = 16$). Appendix F

compares the two readouts on one slot.

H.8 Factored labels

At $d = 8$ the affine dimension of the log-conditionals is 8, 7 and 5 for $K = 1, 2, 4$ under the mean-field readout, never above d , and strictly above d under the exact readout on the same contexts.

Iso-parameter diagonals. Factoring shrinks the arc factors but not the word-label factors (Appendix B.4), so we compare (d, K) pairs with equal arc budgets (Table 16). With the mean-field readout, 8,256 arc parameters give 283.8 ± 8.3 at (16, 1), 230.6 ± 4.4 at (32, 2) and 230.5 ± 11.0 at (64, 4). At 32,896 the seeds disagree at (32, 1), and (64, 2) gives 220.6 ± 3.2 . The extra label width that factoring allows is worth about 50 perplexity, most of it in the first halving.

Readout penalty. The exact readout’s penalty falls monotonically with K on both diagonals (Table 15). At $K = 1$ the product of K mixtures is one mixture, so the richer family has nothing extra to express; splitting the label variable gives it something. At $K = 4$ the exact readout is also the stable one (± 0.8 and ± 4.6 against ± 11.0 and ± 9.4). The sign change at $K = 4$ (-3.5 and -6.2) is within a seed spread, so we claim only the downward trend across the five measured cells on two budgets.

Fixed width. At $d = 32$, raising K cuts the arc budget to 32,896, 8,256 and 2,080. The exact readout improves throughout (313.0 ± 9.5 , 260.6 ± 1.4 , 253.0 ± 1.7) as 94 % of the arc parameters go. The mean-field readout gives 230.6 ± 4.4 at $K = 2$ and 250.5 ± 7.0 at $K = 4$, and its seeds disagree at $K = 1$. This is the only setting in which the exact readout improves as the arc budget shrinks; at $K = 4$ it matches the mean-field readout with a quarter of its seed spread. At $d = 64$, $K = 1$ (131,328 arc parameters), the mean-field readout collapses on all seeds while the exact one trains to 460.1 ± 14.1 : the two do not fail together. None of this recommends $K = 4$. The best cell is the mean-field model at $d = 32$, $K = 2$, and every cell is well above the best perplexities in Table 1.

H.9 WikiText-2

Every constant is carried over from PTB, although the vocabulary is about three times larger (Table 14). The standardised temperature’s seed spread at $d = 32$ is an order of magnitude tighter

arc budget	$K = 1$	$K = 2$	$K = 4$
8,256	+52.2	+30.0	-3.5
32,896	—	+21.7	-6.2

Table 15: Exact-readout perplexity minus mean-field perplexity at a fixed arc budget, three seeds per cell. The $K = 1$ cell at 32,896 has seeds that disagree and no mean.

fam.	d	K	readout	n	non-emb.	val ppl
P	16	1	exact	3	8,256	336.0 ± 6.7
P	16	1	m. field	3	8,256	283.8 ± 8.3
P	32	2	exact	3	8,256	260.6 ± 1.4
P	32	2	m. field	3	8,256	230.6 ± 4.4
P	64	4	exact	3	8,256	227.0 ± 0.8
P	64	4	m. field	3	8,256	230.5 ± 11.0
P	32	1	exact	3	32,896	313.0 ± 9.5
P	32	1	m. field	3	32,896	<i>251–375</i>
P	64	2	exact	3	32,896	242.3 ± 8.5
P	64	2	m. field	3	32,896	220.6 ± 3.2
P	128	4	exact	3	32,896	217.6 ± 4.6
P	128	4	m. field	3	32,896	223.8 ± 9.4
W	32	4	exact	3	2,080	253.0 ± 1.7
W	32	4	m. field	3	2,080	250.5 ± 7.0
W	64	1	exact	3	131,328	460.1 ± 14.1
W	64	1	m. field	3	131,328	<i>3/3 collapsed</i>

Table 16: Factored labels, three seeds. Family P walks two iso-parameter diagonals (8,256 and 32,896 non-embedding parameters), so d rises with K ; family W adds the remaining fixed-width cells. Family W holds the total width fixed, which is where the prediction of Section 2.6 applies; on family P, d rises with K , so a mean-field gain there comes from label width, not from factoring. Original temperature, $\alpha_Z = 0.25$, $h = 4$. Italic ranges and collapse counts as in the introduction to this appendix.

than the original temperature’s. The gap to the baselines persists: with fewer non-embedding parameters than the causal PT’s 65,664, a causal transformer reaches 179.7 ± 4.3 (50,880 parameters) and a looped one 167.9 ± 0.7 (50,112), two seeds each. At matched d_{model} the looped transformer loses to the standard one at every width. At matched non-embedding budget it wins at about 5×10^4 parameters (167.9 against 179.7) and loses at about 2×10^5 (157.2 against 150.9), the same size-dependent inversion as on PTB.